

SMART CROP PROTECTION AND MANAGEMEMT SYSTEM

A PROJECT REPORT

Submitted by

N LOHITH KUMAR	21781A04C7
N JAGADISWAR	21781A04C8
P POORNA CHANDRA REDDY	21781A04D9
P DEVANAND	21781A04E8
P HEMANTH REDDY	21781A04F4

in partial fulfilment of the requirements for the Award of Degree
of

BACHELOR OF TECHNOLOGY

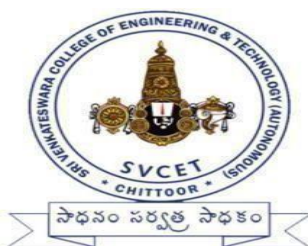
IN

ELECTRONICS AND COMMUNICATION ENGINEERING

Under the guidance of

Mr.K.MURUGANANDAM, M.E

Associate Professor



**SRI VENKATESWARA COLLEGE OF ENGINEERING& TECHNOLOGY
(AUTONOMOUS)**

R.V.S NAGAR, CHITTOOR – 517 127. (A.P).

(Approved by AICTE, New Delhi, Affiliated to JNTUA, Anantapuramu)

(Accredited by NBA, New Delhi AND NAAC A⁺, Bengaluru)

(An ISO 9001:2000 Certified Institution)

APRIL 2025

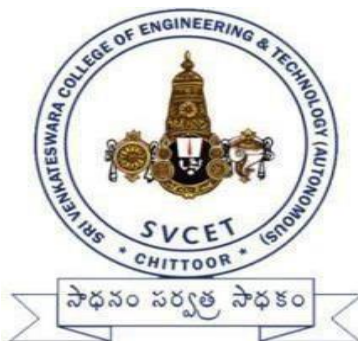
SRI VENKATESWARA COLLEGE OF ENGINEERING& TECHNOLOGY
(AUTONOMOUS)

R.V.S NAGAR, CHITTOOR – 517 127. (A.P)

(Approved by AICTE, New Delhi, Affiliated to JNTUA, Anantapuramu)

Accredited by NBA, New Delhi & NAAC A⁺, Bengaluru)

(An ISO 9001:2000 Certified Institution)



CERTIFICATE

This is to certify that this project entitled “**SMART CROP PROTECTION AND MANAGEMENT SYSTEM**” is a bonafide work done by **N LOHITH KUMAR (21781A04C7)** , **N JAGADISWAR (21781A04C8)** , **P POORNA CHANDRA REDDY (21781A04D9)** , **P DEVANAND (21781A04E8)** , **P HEMANTH REDDY (21781A04F4)** , in partial fulfilment of the requirements for the award of the degree of **BACHELOR OF TECHNOLOGY in ELECTRONICS AND COMMUNICATION ENGINEERING**

SIGNATURE OF THE GUIDE

SIGNATURE OF THE HOD

Viva-Voce Conducted on _____

INTERNAL EXAMINER

EXTERNAL EXAMINER

Acknowledgements

A Grateful thanks to **Dr.R.Venkataswamy**,Chairman,Sri Venkateswara College of Engineering and Technology for providing education in their esteemed institution.

We, wish to record our deep sense of gratitude and profound thanks to our beloved Vice Chairman, **Sri. R.V. Srinivas** for his valuable support throughout the course.

We, express our sincere thanks to **Dr. M. Mohan Babu**, our beloved principal for his encouragement and suggestions during the course of study.

We, wish to convey our gratitude and express our sincere thanks to **Dr.D.Srihari**, Ph.D, Head of the Department, Electronics and Communication Engineering, for giving us his inspiring guidance in undertaking our project report.

We, express our sincere thanks to the Project Guide **Mr.K.Muruganandam** ,M.E, Associate Professor, Department of Electronic and Communication Engineering, for his keen interest, stimulating guidance, constant encouragement with our work during all stages, to bring this project into fruition.

We, wish to convey our gratitude and express our sincere thanks to all Project Review Committee members for their support and cooperation rendered for successful submission of our project work.

Finally, we would like to express our sincere thanks to all teaching,nonteaching faculty members, our parents, and friends and for all those who have supported us to complete the project work successfully.

N LOHITH KUMAR	21781A04C7
N JAGADISWAR	21781A04C8
P POORNA CHANDRA REDDY	21781A04D9
P DEVANAND	21781A04E8
P HEMANTH REDDY	21781A04F4

TABLE OF CONTENTS

INDEX

CHAPTER NO	CONTENTS	PAGE NO
	ABSTRACT	
	LIST OF FIGURES	
	LIST OF TABLES	
1	INTRODUCTION	01
	1.1 Introduction	01
	1.2 Embedded System	01
	1.3 Characteristics	03
	1.4 Microprocessor	03
	1.5 Comparision Between MP And MC	05
2	LITERATURE SURVEY	07
3	OVERVIEW OF THE PROJECT	08
	3.1 Introduction	08
	3.2 Existing System	09
	3.3 Proposed System	09
	3.3.1 Hardware Components	11
	3.1.2 Software Components	11
4	TECHNOLOGY USED	12
	4.1 Introduction of Iot Technology	12

	4.1.1 The Vision	13
	4.2 Definition Of IOT	15
	4.3 24Hours In Feature With IOT	16
	4.4 Architecture Of IOT	17
	4.5 IOT Elements	19
	4.6 Radio Frequency Identification	19
	4.7 Wireless Sensor Networks	19
	4.8 Applications	21
	4.9 IOT In 2016	27
	4.10 Cloud Computing	28
5	HARDWARE IMPLEMENTATION	30
	5.1 Power Supply	30
	5.2 Arduino Uno	32
	5.3 ESP32 Cam	50
	5.4 Ultrasonic Sensor	52
	5.5 Soil Moisture Sensor	53
	5.6 SMTP	54
	5.7 WI-FI Module	54
	5.8 LCD	56
	5.9 Buzzer	58
6	SOFTWARE IMPLEMENTATION	59
	6.1 Arduino Software	59

	6.2 Proteus	66
7	RESULT	69
	7.1 OPERATION	70
8	ADVANTAGES AND APPLICATIONS	71
	8.1 Advantages	71
	8.2 Applications	72
9	CONCLUSION AND FEATURE SCOPE	73
10	REFERENCE	74

LIST OF FIGURES

FIGURE NO	NAME OF THE FIGURE	PAGE NO
1.1	Microprocessor	04
1.2	Microcontroller	04
3.1	Proposed System	10
4.1	IOT Technology	12
5.1	RPS	31
5.2	Capacitor	32
5.3	Structure Of Arduino Board	33
5.4	Arduino Board	34
5.5	Pin Configuration Of Atmega328	37
5.6	AVR CPU Core	41
5.7	AVR Core Architecture	42
5.8	ESP32 Cam	51
5.9	Ultrasonic Sensor	52
5.10	Soil Moisture Sensor	53
5.11	Wi-Fi Module	55
5.12	LCD	56
5.13	Buzzer	58
7.1	Hardware Connection Of The Kit	69
7.2	Output Image	69
7.3	Output Values	69

LIST OF TABLES

SL.NO	TABLE	PAGE NO
1	Table Stack Pointer instructions	48
2	Stack Pointer High and Low Register	48

ABSTRACT

In India, an important growing sector is an agriculture. The major challenges in agriculture are crop productivity, soil nutrient, crop monitoring, etc. This project reviews the internet of things (IoT) based smart agriculture system using IOT. The main purpose of this work is to improve the efficiency of the existing irrigation system and to reduce the human intervention for the complete automation of the system. The proposed system consists of ARDUINO, ULTRASONIC Sensor, ESP32 Cam. ARDUINO is the main controlling unit that can control the operation of ULTRASONIC Sensor, ESP32 Cam. In a smart monitoring system, the pi camera captures the IMAGE and transfers it to the cloud through ARDUINO. All this sensor data is stored in a Internet cloud. So that we can view it from a remote location using Free mobile android application. All this sensor data we can also view on the telegram application using bot AP

INTRODUCTION

1.1 INTRODUCTION:

Micro controller are widely used in Embedded Systems products. An Embedded product uses the microprocessor (or microcontroller) to do one task & one task only. A printer is an example of Embedded system since the processor inside it perform one task only namely getting the data and printing it. Although microcontroller is preferred choice for many Embedded systems, there are times that a microcontroller is inadequate for the task. For this reason, in recent years many manufactures of general-purpose microprocessors such as INTEL, Motorola, AMD & Cyrix have targeted their microprocessors for the high end of Embedded market. One of the most critical needs of the embedded system is to decrease power consumptions and space. This can be achieved by integrating more functions into the CPU chips. All the embedded processors have low power consumptions in additions to some forms of I/O, ROM all on a single chip. In higher performance Embedded system, the trend is to integrate more & more function on the CPU chip & let the designer decide which feature he/she wants to use.

1.2 EMBEDDED SYSTEM:

Physically, embedded systems range from portable devices such as digital watches and MP3 Player's to large stationary installations like traffic lights factory controllers, or the systems controlling nuclear power plants. Complexity varies from low, with a single microcontroller chip, to very high with multiple units, peripherals and networks mounted inside a large chassis or enclosure

Computers share some elements with embedded systems such as the operating systems and microprocessors which power them but are not truly embedded systems, because they allow different applications to be loaded and peripherals to be connected.

Embedded systems span all aspects of modern life and there are many examples of their use. Telecommunications systems employ numerous embedded systems from telephone switches for the network to mobile phones at the end-user. Computer networking uses dedicated routers and network bridges to route data.

EXAMPLES OF EMBEDDED SYSTEM:

- Automated teller machines (ATMS).
- Integrated system in aircraft and missile.
- Cellular telephones and telephonic switches.
- Computer network equipment, including routers, time servers and firewalls
- Computer printers, Copiers.
- Disk drives (floppy disk drive and hard disk drive)
- Engine controllers and antilock brake controllers for automobiles.
- Home automation products like thermostat, air conditioners, sprinklers and security monitoring system.
- Household appliances including microwave ovens, washing machines, TV sets, DVD players/recorders.
- Medical equipment.
- Measurement equipment such as digital storage oscilloscopes, logic analyzers and spectrum analyzers.
- Multimedia appliances: internet radio receivers, TV set top boxes.
- Small hand-held computer with P1, M5 and other applications.
- Programmable logic controllers (PLC's) for industrial automation and monitoring.
- Stationary video game controllers.

1.3 CHARACTERISTICS:

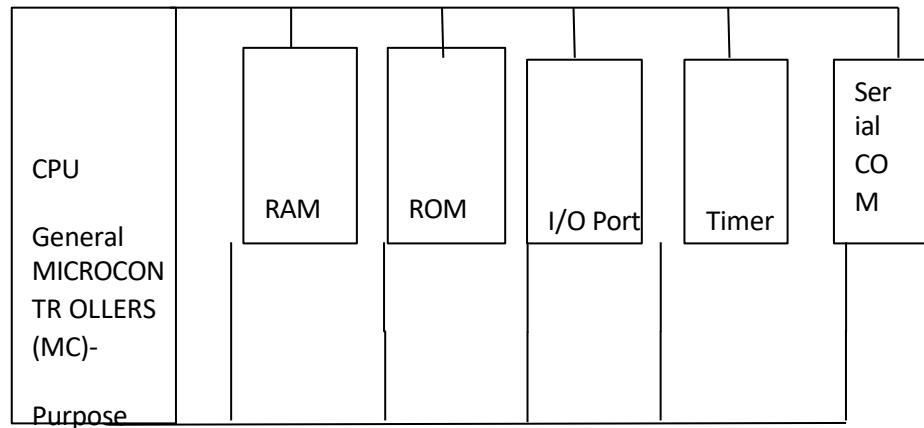
Embedded systems are designed to do some specific tasks, rather than be a general-purpose computer for multiple tasks. Some also have real-time performance constraints that must be met, for reasons such as safety and usability; others may have low or no performance requirements, allowing the system hardware to be simplified to reduce costs.

Embedded systems are not always standalone devices. Many embedded systems consist of small, computerized parts within a larger device that serves a more general purpose. For example, The Gibson Robot Guitar features an embedded system for tuning the strings, but the overall purpose of the Robot Guitar is, of course, to play music. Similarly, an embedded system in an auto mobile provides a specific function as a subsystem of the car itself.

The software written for embedded systems is often called firmware and is usually stored in read-only memory or Flash memory chips rather than a disk drive. It often runs with limited computer hardware resources: small or no keyboard, screen, and little memory.

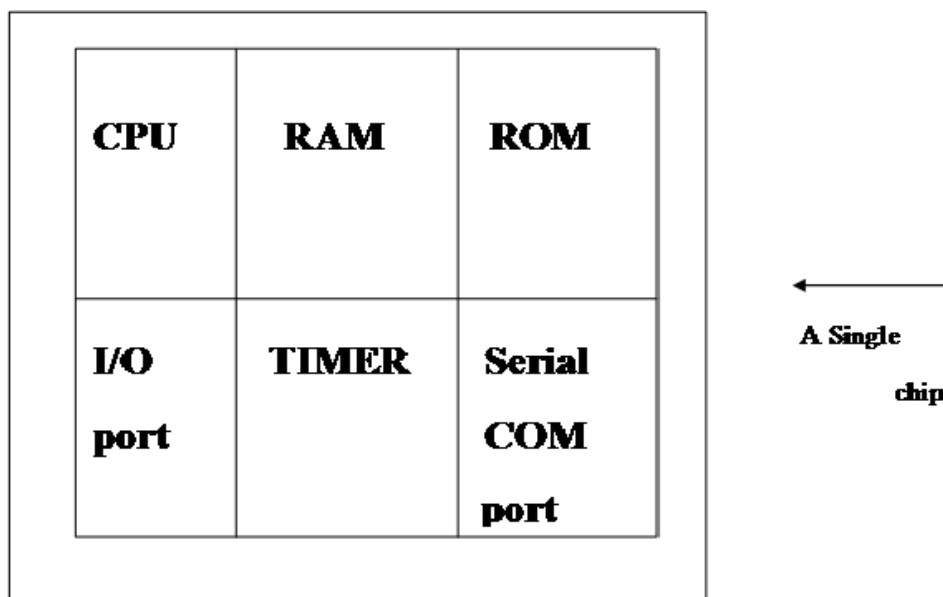
1.4 MICRO PROCESSOR (MP):

A microprocessor is a general-purpose digital computer central processing unit (CPU). Although popularly known as a “computer on a chip” is in no sense a complete digital computer. The block diagram of a microprocessor CPU is shown, which contains an arithmetic and logical unit (ALU), a program counter (PC), a stack pointer (SP), some working registers, a clock timing circuit, and interrupt circuits.

**Fig 1.1 Microprocessor**

MICROCONTROLLER(MC):

Figure shows the block diagram of a typical microcontroller. The design incorporates all of the features found in micro-processor CPU: ALU, PC, SP, and registers. It also added the other features needed to make a complete computer: ROM, RAM, parallel I/O, serial I/O, counters, and clock circuit.

**Fig 1.2 Microcontroller**

1.5 COMPARISION BETWEEN MICROPROCESSOR AND MICROCONTROLLER

The microprocessor must have many additional parts to be operational as a computer whereas microcontroller requires no additional external digital parts.

1. The prime use of microprocessor is to read data, perform extensive calculations on that data and store them in the mass storage device or display it. The prime functions of microcontroller is to read data, perform limited calculations on it, control its environment based on these data. Thus the microprocessor is said to be general-purpose digital computers whereas the microcontroller are intend to be special purpose digital controller.

2. Microprocessor need many opcodes for moving data from the external memory to the CPU, microcontroller may require just one or two, also microprocessor may have one or two types of bit handling instructions whereas microcontrollers have many.

PERIPHERALS:

Embedded Systems talk with the outside world via peripherals such as

- Serial Communication Interfaces (SCI): RS-232, RS-422, RS-485etc
- Synchronous Serial Communication Interface: I2C, JTAG, SPI, SSC and ESSI
- Universal Serial Bus (USB)
- Networks: Ethernet, Controller Area Network, LAN networks, etc
- Timers: PLL(s), Capture/Compare and Time Processing Units
- Discrete IO: aka General Purpose Input/output (GPIO)
- Analog to Digital/Digital to Analog (ADC/DAC)

TOOLS:

As for other software, embedded system designers use compilers, assemblers, and debuggers to develop embedded system software. However, they may also use some more specific tools:

- Utilities to add a checksum or CRC to a program, so the embedded system can check if the program is valid.
- For systems using digital signal processing, developers may use a math workbench such as MATLAB, Simulink, Mathcad, or Mathematica to simulate the mathematics. They might also use libraries for both the host and target which eliminates developing DSP routines as done in DSP nano RTOS and Unison Operating System.
- Custom compilers and linkers may be used to improve optimization for the particular hardware.
- An embedded system may have its own special language or design tool, or add enhancements to an existing language such as Forth or Basic.
- Another alternative is to add a Real-time operating system or Embedded operating system, which may have DSP capabilities like DSP Nano RTOS.

LITERATURE SURVEY

Redmon, J & Farhadi, A (2018): Mentioned for their work on YOLOv3, an object detection system relevant to real-time intrusion detection in agriculture. Their work would likely be discussed in surveys focusing on AI and computer vision in crop protection.

Viola, P. & Jones, M (2001): Their object detection framework is referenced as being applicable to real-time detection of intruders in farmland, suggesting their methods are relevant to surveys on this topic.

Korche, M., Tokse, S., Shirbhate, S., & Jolhe, S. P. (2021): Authored a study proposing a microcontroller-based smart crop protection system, which would likely be cited in relevant surveys.

Talavera et al. (2017 - Based on similar searches): While not explicitly dated in these snippets, other searches indicate Talavera et al. published a systematic literature review of IoT applications in environmental and agricultural industrial fields from 2006 to 2016.

OVERVIEW OF THE PROJECT

3.1 INTRODUCTION

Water scarcity is a big problem for farming due to droughts and not enough rain. In the past, farmers used basic methods like buckets and canals to water their crops, which needed a lot of work and led to under- or over-irrigation, which can damage crops and waste water. But now, Smart irrigation systems use technology to deliver the right amount of water to crops at the right time. They can help to conserve water and improve crop yields. Machine learning can be used to improve the accuracy of smart irrigation systems. Machine learning models can take into account various factors, such as soil moisture, temperature, and weather, to predict when and how much to irrigate. They use smart irrigation systems that use technology like the Internet of Things (IoT) to control water distribution. These systems are better at saving water and helping crops grow well. These smart systems use things like sprinklers and pipes to give the right amount of water to the plants when they need it. They also use machine learning to know exactly how much water the plants should get. Different types of machine learning techniques, such as random forest algorithm, convolutional neural network, support vector machine, logistic regression are used to make these systems work better. The experimental findings show that the random forest algorithm achieves the highest accuracy of 99.98% in classifying device status, followed by the convolutional neural network with an accuracy of 98.23%. The support vector machine algorithm achieves an accuracy of 90.21%, while logistic regression lags behind with an accuracy of 71.76%. They use data collected from sensors to understand when and how much to water the crop needed. This data can be in different forms like numbers. They also use machine learning methods like logistic regression and random forests to decide if the smart irrigation system is working properly or not. If the system can't know if it's on or off, it might cause problems for the crops. So, we use these algorithms to make sure everything works smoothly in smart farming. The main goal is to make these systems really good at smart irrigation. Researchers have been working on different ways to make farming smarter. I or liam and team in 2020 came up with a

clever idea. They used artificial neural networks to understand the soil's nutrients through data from smart soil sensors. Their approach was pretty accurate, ranging from 81.33% to 97.13%.

3.2 EXISTING SYSTEM

Consist of only one moisture sensor which does not cover the whole field area. DC motor is used which is directly connected to relay and power supply is given from Arduino board. When the moisture sensor detect the water levels the ON/OFF of motor takes place. When the water is low i.e. low moisture the motor is turned On, and when the water level is high i.e. high moisture the motor is turned off. LCD shows the present state of moisture sensor. There is no wireless transmission of data to the farmer.

3.3 PROPOSED SYSTEM

The concept of smart agriculture is gaining popularity, supported by Machine Learning and Internet of Things sensors that collect information from agricultural areas and respond to human inputs. Smart farming systems that use remote sensing can increase production, reduce waste, and efficiently manage a wider range of resources. The development of Smart Agriculture Systems using IOT and ML not only helps to improve the farming quality but also saves farmer's lot of time. In our system this will be accomplish with the use of soil moisture sensor is connected to Arduino Uno.

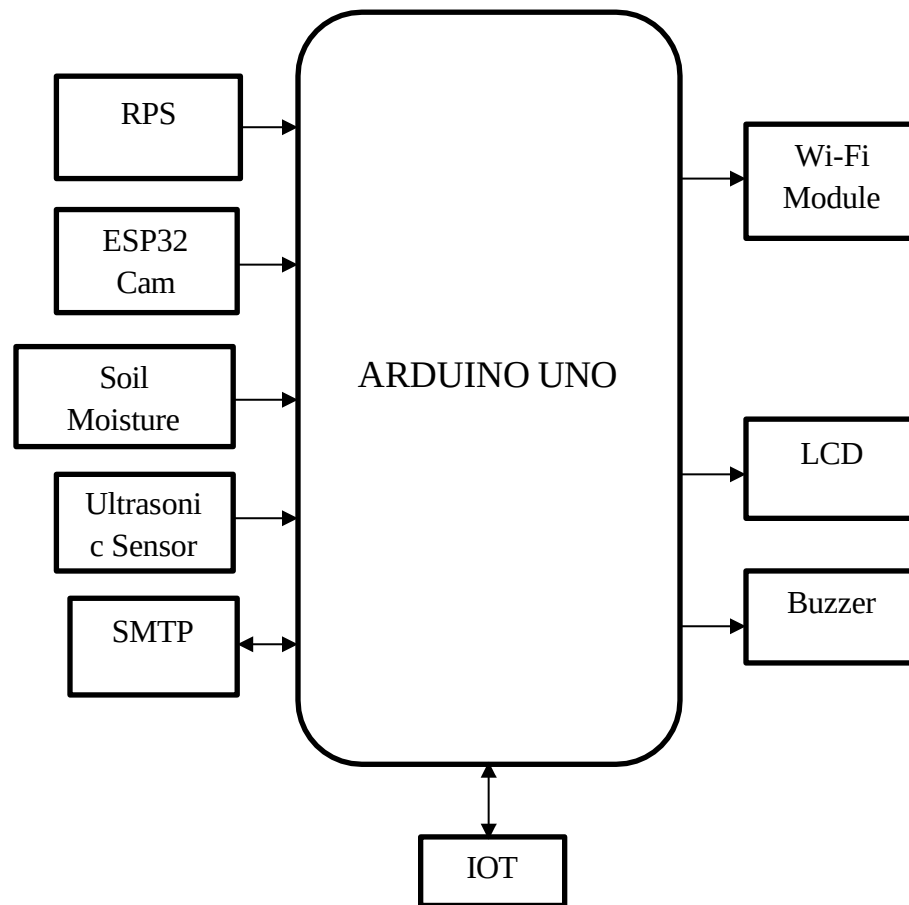


Fig 3.1 Proposed System

3.3.2 HARDWARE COMPONENTS

- Power Supply
- Arduino UNO
- ESP32 cam
- Ultrasonic Sensor
- Soil Moisture Sensor
- SMTP
- Wi-Fi Module
- LCD
- Buzzer

3.3.3 SOFTWARE COMPONENTS

- Arduino IDE
- Proteus

TECHNOLOGY USED

TECHNOLOGIES:

IOT

INTRODUCTION OF IOT TECHNOLOGY

1. The term "Internet of Things" has come to describe a number of technologies and research disciplines that enable the Internet to reach out into the real world of physical objects.
2. The Internet of Things, also called The Internet of Objects, refers to a wireless network between objects.
3. From any time, any place connectivity for any one, we will now have connectivity for anything!

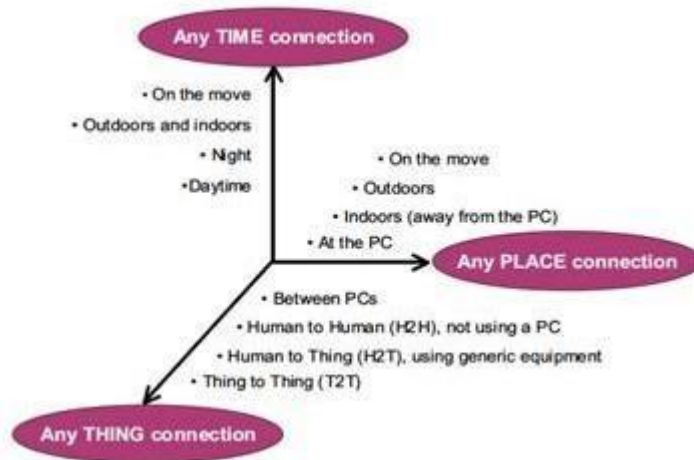


Fig4.1 IOT Technology

4.1.1 THE VISION

To improve human health and well-being is the ultimate goal of any economic, technological and social development. The rapid rising and aging of population are one of the macro powers that will transform the world dramatically, it has caused great pressure to food supply and healthcare systems all over the world, and the emerging technology break through of the Internet-of-Things (IoT) is expected to offer promising solutions (National Information Council 2008). Therefore, the application of IoT technologies for the food supply chain (FSC) (so-called Food-IoT) and in-home healthcare (IHH) (so-called Health-IoT 1) have been naturally highlighted in the strategic research road maps (European Commission Information Society 2009). To develop practically usable technologies and architectures of IoT for these two applications is the final target of this work. The phrase "Internet of Things" (IoT) was coined at the beginning of the 21st century by the MIT Auto-ID Center with special mention to Kevin Ashton (Ashton 2009) and David L. Brock (Brock 2001). As a complex cyber-physical system, the IoT

Integrates all kinds of sensing, identification, communication, networking, and informatics devices and systems, and seamlessly connects all the people and things upon interests, so that anybody, at any time and any place, through any device and media, can more efficiently access the information of any object and any service (ITU 2005, European Commission Information Society 2008 and 2009). "Ubiquitous" is the distinct feature of IoT technologies, so the IoT is often related to ubiquitous identification (Sheng et al. 2010), ubiquitous sensing (ITU-T, 2008), ubiquitous computing (Fried Ewald and Raabe 2011), ubiquitous intelligence (Zheng et al. 2008), etc. As shown in Figure 1-1, a vivid description of this vision has been illustrated in a report by The Economist in 2007 (The Economist 2007).

A vivid description of the vision of Internet-of-Things (Authorized by Jon Berkeley) The impact caused by the IoT to human life will be as huge as the internet has caused in the past decades, so the IoT is recognized as "the next of internet". A part of the enabling technologies are sensors and actuators, Wireless Sensor Network

(WSN), Intelligent and Interactive Packaging (I2Pack), real-time embedded system, Micro electro mechanical Systems (MEMS), mobile internet access, cloud computing, Radio Frequency Identification (RFID), Machine-to-Machine (M2M) communication, human machine interaction (HMI), middleware, Service Oriented Architecture(SOA),Enterprise Information System(EIS), data mining, etc. With various descriptions from various viewpoints, the IoT has become the new paradigm of the evolution of information and communication technology (ICT).

CHARACTERISTICS FOR INTERNET OF THINGS:

- Event driven
- Ambient intelligence
- Flexible structure
- Semantic sharing
- Complex access technology

Anyone who says that the Internet has fundamentally changed society may be right, but at the same time, the greatest transformation actually still lies ahead of us. Several new technologies are now converging in a way that means the Internet is on the brink of a substantial expansion as objects large and small get connected and assume their own web identity.

Following on from the Internet of computers, when our servers and personal computers were connected to a global network ,and the Internet of mobile telephones, when it was the turn of telephones and other mobile units, the next phase of development is the Internet of things, when more or less anything will be connected and managed in the virtual world. This revolution will be the Net's largest enlargement ever and will have sweeping effects on every industry — and all of our everyday lives.

Smart connectivity with existing networks and context-aware computation using network resources is an indispensable part of IoT. With the growing presence of Wi-Fi and 4G-LTE wireless Internet access, the evolution towards ubiquitous

information and communication networks is already evident. However, for the Internet of Things vision to successfully emerge, the computing paradigm will need to go beyond traditional mobile computing scenarios that use smart phones and portables, and evolve into connecting everyday existing objects and embedding intelligence into our environment. For technology to disappear from the consciousness of the user, the Internet of Things demands :a shared understanding of the situation of its users and their appliances, software architectures and pervasive communication networks to process and convey the contextual information to where it is relevant, and the analytics tools in the Internet of Things that aim for autonomous and smart behavior. With these three fundamental grounds in place, smart connectivity and context-aware computation can be accomplished.

A radical evolution of the current Internet into a Network of interconnected objects that not only harvests information from the environment (sensing) and interacts with the physical world (actuation/ command/control), but also uses existing Internet standards to provide services for information transfer, analytics, applications and communications. Fueled by the prevalence of devices enabled by open wireless technology such as Bluetooth, radio frequency identification (RFID), Wi-Fi, and telephonic data services as well as embedded sensor and actuator nodes, IoT has stepped out of its infancy and is on the verge of transforming the current static Internet into a fully integrated Future Internet.

The Internet revolution led to the interconnection between people at an unprecedented scale and pace. The next revolution will be the interconnection between object to create smart environment. Only in 2011 did the number of interconnected devices on the planet overtake the actual number of people. Currently there are 9 billion interconnected devices and it is expected to reach 24 billion devices by 2020. According to the GSMA, this amounts to \$1.3 trillion revenue opportunities for mobile network operators alone spanning vertical segments such as health, automotive, utilities and consumer electronics.

4.2 DEFINITION OF INTERNET OF THINGS (IoT)

“Today computers and, therefore, the Internet are almost wholly

dependent on human beings for information. Nearly all of the roughly 50 [petabytes](#)(a petabyte is 1,024 terabytes) of data available on the Internet were first captured and created by human beings— by typing, pressing a record button, taking a digital picture, or scanning a bar code. Conventional diagrams of the Internet ... leave out the most numerous and important routers of all - people. The problem is, people have limited time, attention and accuracy—all of which means they are not very good at capturing data about things in the real world. And that big deal. We're physical, and so is our environment ... You can't eat bits, burn them to stay warm or put them in your gas tank. Ideas and information are important, but things matter much more. Yet today's information technology is so dependent on data originated by people that our computers know more about ideas than things. If we had computers that knew everything there was to know about things—using data they gathered without any help from us—we would be able to track and count everything, and greatly reduce waste, loss and cost. We would know when things needed replacing, repairing or recalling, and whether they were fresh or past their best. The Internet of Things has the potential to change the world, just as the Internet did. Maybe even more so”.

“Things are active participants in business, information and social processes where they are enabled to interact and communicate among themselves and with the environment by exchanging data and information sensed about the environment, while reacting autonomously to the real/physical world events and influencing it by running processes that trigger actions and create services with or without direct human intervention.”

“The Internet of Things represents an evolution in which objects are capable of interacting with other objects. Hospitals can monitor and regulate pacemakers’ long distance, factories can automatically address production line issues and hotels can adjust temperature and lighting according to a guest's preferences, to name just a few examples.”

4.3 24 HOURS IN FUTURE WITH INTERNET OF THINGS

How many devices do you currently own? Three, four or more? A laptop, a tablet, a smart phone or even a smart watch and a smart band or the new Nike smart

shoes maybe? It goes without saying that technology has become an indispensable part of our lives.

Here is how 24 hours would like in the future with IoT-

I wouldn't need to set an alarm for the next day. My smart phone will pick up the time of my meeting from my digital calendar will connect to estimate the real time traffic and predict my travelling time to the meeting venue will analyze how much time I usually take to get ready AND finally it will calculate how early I need to wake up!

My smart heater would know that I am up and it will adjust the water temperature according to my preference.

When I'll be ready to leave the house, the smart refrigerator will tell me that there is milk and fruits for my breakfast and I shouldn't eat cheese today because I didn't exercise yesterday. It is so smart that if any food item has finished or expired- it will connect to the supermarket on the internet, order my groceries and I would be able to pay sitting from anywhere anytime through my mobile wallet.

When I am finished eating breakfast, my smart car will turn the AC on so that as soon as I start driving there is a favorable temperature in the car and smart radio will be automatically turned to my favorite radio station. When let's say I get back from a yoga class, my smart clothing will know that I am sweating and this data will be sent to my smart home temperature system which will adjust the room temperature as soon as I enter the house to make me feel comfortable.

These devices are so smart that they would be collecting data from my physical movements and will monitor my activities and behavior to do predictive analysis of my routine and preferences. You would be amazed at how many such smart devices already exist or are going to be available in the near future to make up a powerful Internet of Things.

4.4 ARCHITECTURE OF INTERNET OF THINGS

Architecture of internet Of Things contains basically 4 layers:

Application Layer

1. Gateway and the network layer
2. Management Service layer
3. Sensor layer

APPLICATION LAYER:

- Lowest Abstraction Layer
- With sensors we are creating digital nervous system.
- Incorporated to measure physical quantities
- Inter connects the physical and digital world
- Collects and process the real time information

GATEWAY AND THE NETWORK LAYER:

- Robust and High-performance network in a structure
- Supports the communication requirements for latency, band width or security
- Allows multiple organizations to share and use the same network independently

MANAGEMENT LAYER:

- Capturing of periodic sensory data
- Data Analytics (Extracts relevant information from massive amount of raw data)
- Streaming Analytics (Process real time data)
- Ensures security and privacy of data.

SENSOR LAYER:

- Provides a user interface for using IoT.
- Different applications for various sectors like Transportation, Health care, Agriculture, Supply chains, Government, Retail etc.

4.5 INTERNET OF THINGS ELEMENTS

There are three IoT components which enables seamless:

- a) Hardware madeup of sensors, actuators and embedded communication hardware
- b) Middleware on demand storage and computing tools for data analytics and
- c) Presentation—novel easy to understand visualization and interpretation tools which canbe widely accessed on different platforms and which can be designed for different applications

4.6 RADIO FREQUENCY IDENTIFICATION(RFID)

RFID technology is a major breakthrough in the embedded communication paradigm which enables design of microchips for wireless data communication. They help in the automatic identification of anything they are attached to acting as an electronic barcode. The passive RFID tags are not battery powered and they use the power of the reader's interrogation signal to communicate the ID to the RFID reader. This has resulted in many applications particularly in retail and supply chain management. The applications can be found in transportation (replacement of tickets, registration stickers) and access control applications as well. The passive tags are currently being used in many bank cards and road toll tags which are among the first global deployments. Active RFID readers have their own battery supply and can instantiate the communication. Of the several applications, the main application of active RFID tags is in port containers for monitoring cargo.

4.7 WIRELESS SENSOR NETWORKS (WSN)

Recent technological advances in low power integrated circuits and wireless communication shave made available efficient, low cost, low power miniature devices

for use in remote sensing applications. The combination of these factors has improved the viability of utilizing a sensor network consisting of a large number of intelligent sensors, enabling the collection, processing, analysis and dissemination of valuable information, gathered in a variety of environments. Active RFID is nearly the same as the lower end WSN nodes with limited processing capability and storage. The scientific challenges that must be overcome in order to realize the enormous potential of WSNs are substantial and multi disciplinary in nature. Sensor data are shared among sensor nodes and sent to a distributed or centralized system for analytics. The components that make up the WSN monitoring network include:

- a) WSN hardware—Typically a node (WSN core hardware) contains sensor interfaces, processing units, transceiver units and power supply. Almost always, they comprise of multiple A/D converters for sensor interfacing and more modern sensor nodes have the ability to communicate using one frequency band making them more versatile.
- b) WSN communication stack—The nodes are expected to be deployed in an ad-hoc manner for most applications. Designing an appropriate topology, routing and MAC layer is critical for the scalability and longevity of the deployed network. Nodes in a WSN need to communicate among themselves to transmit data in single or multi-hop to a base station. Node drop outs, and consequent degraded network lifetimes, are frequent.
- c) WSN Middle ware—A mechanism to combine cyber infrastructure with a Service Oriented Architecture (SOA) and sensor networks to provide access to heterogeneous sensor resources in a deployment independent manner. This is based on the idea of isolating resources that can be used by several applications. A platform-independent middleware for developing sensor applications is required, such as an Open Sensor Web Architecture.
- d) Secure Data aggregation—An efficient and secure data aggregation method is required for extending the lifetime of the network as well as ensuring reliable data collected from sensors. Node failures are a common characteristic of WSNs, the network topology should have the capability to

heal itself. Ensuring security is critical as the system is automatically linked to actuators and protecting the systems from intruders becomes very important.

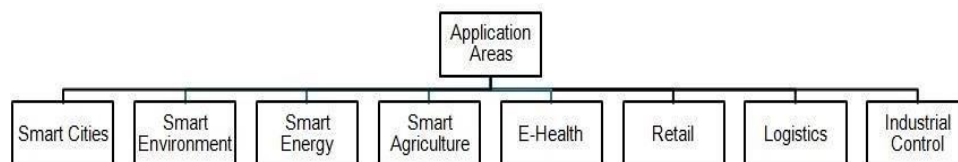
4.8 APPLICATIONS:

There are several application domains which will be impacted by the emerging Internet of Things. The applications can be classified based on the type of network availability, coverage, scale, heterogeneity, repeatability, user involvement and impact.

We categorize the applications into four application domains:

- 1) Personal and Home
- 2) Enterprise
- 3) Utilities
- 4) Mobile.

There is a huge crossover in applications and the use of data between domains. For instance, the Personal and Home IoT produces electricity usage data in the house and makes it available to the electricity (utility) company which can in turn optimize the supply and demand in the Utility IoT. The internet enables sharing of data between different service providers in a seamless manner creating multiple business opportunities.



PERSONAL AND HOME

The sensor information collected is used only by the individuals who directly own the network. Usually, Wi-Fi is used as the backbone enabling higher bandwidth data (video) transfer as well as higher sampling rates (Sound).

Ubiquitous healthcare has been envisioned for the past two decades. IoT gives a perfect platform to realize this vision using body area sensors and IoT back end to upload the data to servers. For instance, a Smart phone can be used for communication along with several interfaces like Bluetooth for interfacing sensors measuring physiological parameters. So far, there are several applications available for Apple iOS, Google Android and Windows Phone operating systems that measure various parameters. However, it is yet to be centralized in the cloud for general physicians to access the same.

An extension of the personal body area network is creating a home monitoring system for elderly care, which allows the doctor to monitor patients and the elderly in their homes thereby reducing hospitalization costs through early intervention and treatment.

Control of home equipment such as air conditioners, refrigerators, washing machines etc., will allow better home and energy management. This will see consumers become involved in the IoT revolution in the same manner as the Internet revolution itself.

Social networking is set to undergo another transformation with billions of inter connected objects. An interesting development will be using a Twitter like concept where individual 'Things' in the house can periodically tweet the readings which can be easily followed from anywhere creating a Tweetot. Although this provides a common framework using cloud for information access, a new security paradigm will be required for this to be fully realized.

ENTERPRIZE

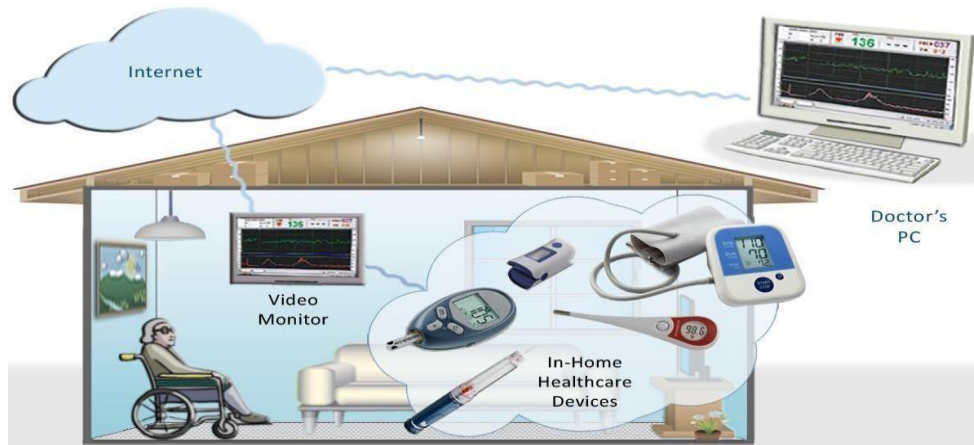
We refer to the 'Network of Things' within a work environment as an enterprise-based application. Information collected from such networks are used only by the owners and the data may be released selectively. Environmental monitoring is the first common application which is implemented to keep track of the number of occupants and manage the utilities within the building (e.g., HVAC, lighting).

Sensors have always been an integral part of the factory setup for security,

automation, climate control, etc. This will eventually be replaced by a wireless system giving the flexibility to make changes to the setup whenever required. This is nothing but an IoT subnet dedicated to factory maintenance.

One of the major IoT application areas that is already drawing attention is Smart Environment IoT. There are several test beds being implemented and many more planned in the coming years. Smart environment includes subsystems and the characteristics from a technological perspective.

These applications are grouped according to their impact areas. This includes the effect on citizens considering health and well-being issues; transport in light of its impact on mobility, productivity, pollution; and services in terms of critical community services managed and provided by local government to city inhabitants.



Utilities

The information from the networks in this application domain is usually for service optimization rather than consumer consumption. It is already being used by utility companies (smart meter by electricity supply companies) for resource management in order to optimize cost vs. profit. These are made up of very extensive networks (usually laid out by large organization on a regional and national scale) for monitoring critical utilities and efficient resource management. The backbone network used can vary between cellular, Wi-Fi and satellite communication.

Smart grid and smart metering are another potential IOT application which is being implemented around the world. Efficient energy consumption can be achieved

by continuously monitoring every electricity point within a house and using this information to modify the way electricity is consumed. This information at the city scale is used for maintaining the load balance within the grid ensuring high quality of service.

Video based IOT, which integrates image processing, computer vision and networking frame works, will help develop a new challenging scientific research area at the intersection of video, infrared, microphone and network technologies. Surveillance, the most widely used camera network applications, helps track targets, identify suspicious activities, detect left luggage and monitor unauthorized access.

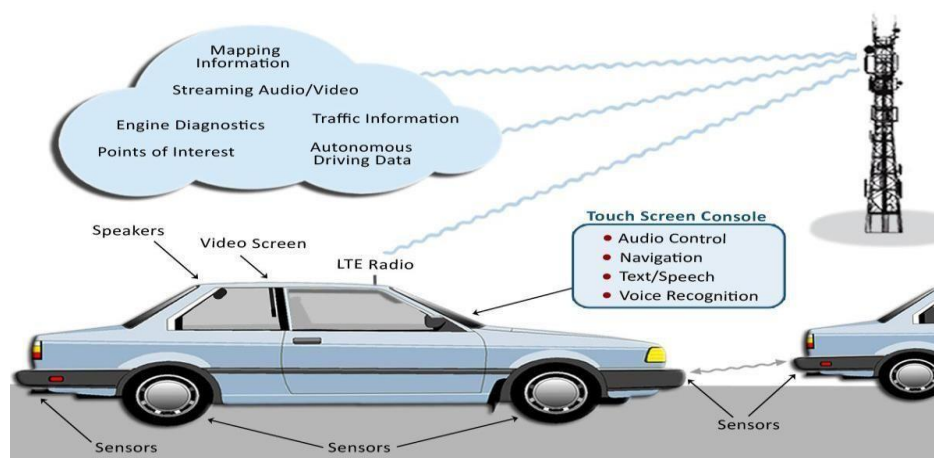
Water network monitoring and quality assurance of drinking water is another critical application that is being addressed using IOT. Sensors measuring critical water parameters are installed at important locations in order to ensure high supply quality. This avoids accidental contamination among storm water drains, drinking water and sewage disposal. The same network can be extended to monitor irrigation in agricultural land. The network is also extended for monitoring soil parameters which allows informed decision-making concerning agriculture.

MOBILE:

Smart transportation and smart logistics are placed in a separate domain due to the nature of data sharing and backbone implementation required. Urban traffic is the main contributor to traffic noise pollution and a major contributor to urban air quality degradation and greenhouse gas emissions. Traffic congestion directly imposes significant costs on economic and social activities in most cities. Supply chain efficiencies and productivity, including just-in-time operations, are severely impacted by this congestion causing freight delays and delivery schedule failures. Dynamic traffic information will affect freight movement, allow better planning and improved scheduling. The transport IoT will enable the use of large scale WSNs for online monitoring of travel times, origin– destination (O–D) route choice behavior, queue lengths and air pollutant and noise emissions. The IoT is likely to replace the traffic information provided by the existing sensor networks of inductive loop vehicle detectors employed at the intersections of existing traffic control systems. They will

also underpin the development of scenario-based models for the planning and design of mitigation and alleviation plans, as well as improved algorithms for urban traffic control, including multi-objective control systems. Combined with information gathered from the urban traffic control system, valid and relevant information on traffic conditions can be presented to travelers. The prevalence of Bluetooth technology (BT) devices reflects the current IoT penetration in a number of digital products such as mobile phones, car hands-free sets, navigation systems, etc. BT devices emit signals with a unique Media Access Identification (MAC-ID) number that can be read by BT sensors within the coverage area.

Readers placed at different locations can be used to identify the movement of the devices. Complemented by other data sources such as traffic signals, or bus GPS, research problems that can be addressed include vehicle travel time on motorways and arterial streets, dynamic (time dependent) O–D matrices on the network, identification of critical intersections, and accurate and reliable real time transport network state information. There are many privacy concerns by such usages and digital forgetting is an emerging domain of research in IoT where privacy is a concern. Another important application in mobile IoT domain is efficient logistics management. This includes monitoring the items being transported as well as efficient transportation planning. The monitoring of items is carried out more locally, say, within a truck replicating enterprise domain but transport planning is carried out using a large scale IoT network.



CLOUD CENTRIC INTERNET OF THINGS

The vision of IoT can be seen from two perspectives— ‘Internet’ centric and ‘Thing’ centric. The Internet centric architecture will involve internet services being the main focus while data is contributed by the objects. In the object centric architecture, the smart objects take the center stage.

In order to realize the full potential of cloud computing as well as ubiquitous sensing, a combined framework with a cloud at the center seems to be most viable. This not only gives the flexibility of dividing associated costs in the most logical manner but is also highly scalable. Sensing service providers can join the network and offer their data using a storage cloud; analytic tool developers can provide their software tools; artificial intelligence experts can provide their data mining and machine learning tools useful in converting information to knowledge and finally computer graphics designers can offer a variety of visualization tools. Cloud computing can offer these services as Infrastructures, Platforms or Software where the full potential of human creativity can be tapped using them as services.

The new IoT application specific frame work should be able to provide support for:

- a) Reading data streams either from sensors directly or fetch the data from databases.
- b) Easy expression of data analysis logic as functions/operators that process data streams in a transparent and scalable manner on Cloud infrastructures
- c) If any events of interest are detected, out comes should be passed to output streams, which are connected to a visualization program. Using such a frame work, the developer of IoT applications will be able to harness the power of Cloud computing without knowing low-level details of creating reliable and scale applications.

BENEFITS OF INTERNET OF THINGS

- Improved citizen's quality of life

Health care from anywhere Better safety, security and productivity

- New business opportunities

IoT can be used in every vertical for improving the efficiency

Creates new businesses, and new and better jobs

- Economic growth

Billions of dollars in savings and new services

- Better environment

Saves natural resources and trees

Helps in creating a smart, greener and sustainable planet

- Improved competitiveness

Competitive in providing cutting edge products/services

4.9 INTERNET OF THINGS IN 2016

SMART WATCHES

Smart watches broke new ground last year, with the popularity of the devices like the Pebble and the Galaxy Gear. More smart watches making their way in the market with better and at the feasible prices. With Apple's long-anticipated but expected announcement of the iWatch, as the company has been ramping up its sapphire glass production and flexible, wearable watch like patents.

Industry Innovators: Pebble, Meta watch, Samsung Galaxy Gear

THE AUTOMATED HOME

Popular devices like Google's Nest Smart Thermostat and WeMo's electrical outlet controller gained in popularity last year. Since then, numerous home automation IoT technologies have flourished- everything from smart locks to Wi-Fi enabled light bulbs.

Industry Innovators: Nest, Lockitron, Lifi

FITNESS AND HEALTH TRACKING

Last year, health and fitness devices like Nike Fuel band and Jawbone Up were among the most popular IoT gadgets, making large appearance at CES.

Industry Innovators: Fit bit, Nike, Jawbone

CONNECTED RETAIL

Traditional retailer store is struggling to keep up with the growing e-commerce. Thanks to the Internet of Things, innovators have started to breathe new life into the retail experience- offering connected point of sale systems, NFC payments solutions and supply chain software's.

Industry Innovators: Shop keep, Cisco, Placemeter

VIRTUAL AUGMENTED REALITY

Last year Oculus Rift and Google glass made headline in both the virtual and augmented Reality worlds. Oculus was acquired by Facebook for \$2.3 Billion earlier this year and Google glass recently rolled out a one-day sale of its "Explorer Edition".

Industry Innovators: Oculus, Google Glass, Sony

4.10 CLOUD COMPUTING

Integrated IoT and Cloud computing applications enabling the creation of smart environments such as Smart Cities need to be able to:

- a) Combine services offered by multiple stake holders
- b) Scale to support a large number of users in a reliable and decentralized manner.

They need to be able operate in both wired and wireless network environments and deal with constraints such as access devices or data sources with limited power and unreliable connectivity.

The Cloud application platforms need to be enhanced to support:

- a) The rapid creation of applications by providing domain specific

programming tools and environments.

- b) Seamless execution of applications harnessing capabilities of multiple dynamic and heterogeneous resources to meet quality of service requirements of diverse users

HARDWARE IMPLEMENTATION

HARDWARE COMPONENTS ARE

- Power Supply
- Arduino UNO
- ESP32 Cam
- Ultrasonic Sensor
- Soil Moisture Sensor
- SMTP
- Wi-Fi Module
- LCD
- Buzzer

5.1 POWER SUPPLY

In this project we have power supplies with +5V & -5V option normally +5V is enough for total circuit. Another (-5V) supply is used in case of OP amp circuit.

Transformer primary side has 230/50HZ AC voltage whereas at the secondary winding the voltage is step downed to 12/50 Hz and this voltage is rectified using two full wave rectifiers the rectified output is given to a filter circuit to filter the unwanted ac in the signal. After that the output is again applied to a regulator LM7805 (to provide +5v) regulator. Whereas LM7905 is for providing -5V regulation. Z (+12V) circuit is used for stepper motors, Fan and Relay by using LM7812 regulator same process like above supplies).

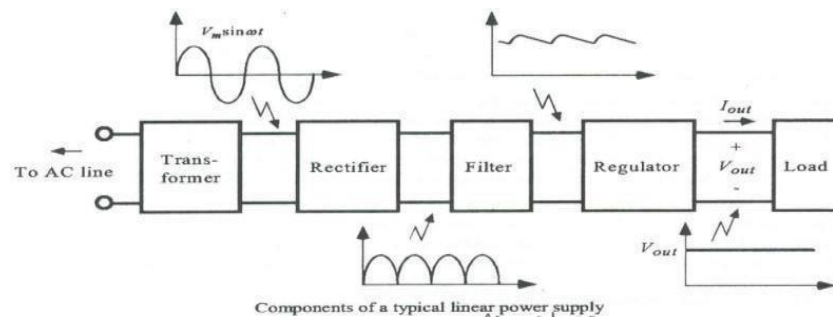


Fig 5.1 RPS

TRANSFORMER

Transformers are used to convert electricity from one voltage to another with minimal loss of power. They only work with AC (alternating current) because they require a changing magnetic field to be created in their core. Transformers can increase voltage (step-up) as well as reduce voltage (step-down).

Alternating current flowing in the primary (input) coil creates a continually changing magnetic field in them on core. This field also passes through the secondary (output) coil and the changing strength of the magnetic field induces an alternating voltage in the secondary coil. If the secondary coils connected to a load the induced voltage will make an induced current flow. The correct term for the induced voltage is 'induced electromotive force' which is usually abbreviated to induced e.m.f.

RECTIFIERS

The purpose of a rectifier is to convert an AC wave form into a DC wave form(OR) Rectifier converts AC current or voltages into DC current or voltage. There are two different rectification circuits, known as 'half-wave' and 'full-wave' rectifiers. Both use components called diodes to convert AC into DC.

FILTERS

A filter circuit is a device which removes the ac component of rectifier output but allows the dc component to the load. The most commonly used filter circuits are capacitor filter, choke input filter and capacitor input filter or pi-filter. We used capacitor filter here.

The capacitor filter circuit is extremely popular because of its low cost, small size, little weight and good characteristics. For small load currents this type of filter is preferred. It is commonly used in transistor radio battery eliminators.

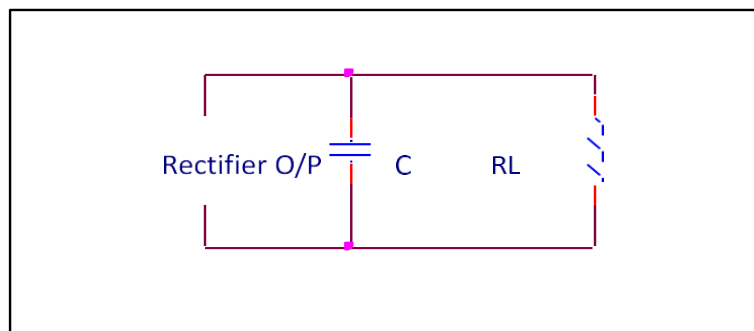


Fig 5.2 Capacitor

5L,X.2 ARDUINO UNO

INTRODUCTION

The Arduino is a family of microcontroller boards to simplify electronic design, prototyping and experimenting for artists, hackers, hobbyists, but also many professionals. People use it as brains for their robots, to build new digital music instruments, or to build a system that lets your house plants tweet you when they're dry. Arduino's (we use the standard Arduino Uno) are built around an ATmega microcontroller — essentially a complete computer with CPU, RAM, Flash memory,

and input/output pins, all on a single chip. Unlike, say, a Raspberry Pi, it's designed to attach all kinds of sensors, LEDs, small motors and speakers, servos, etc. directly to these pins, which can read in or output digital or analog voltages between 0 and 5 volts. The Arduino connects to your computer via USB, where you program it in a simple language (C/C++, similar to Java) from inside the free Arduino IDE by uploading your compiled code to the board. Once programmed, the Arduino can run with the USB link back to your computer, or stand-alone without it — no keyboard or screen needed, just power.

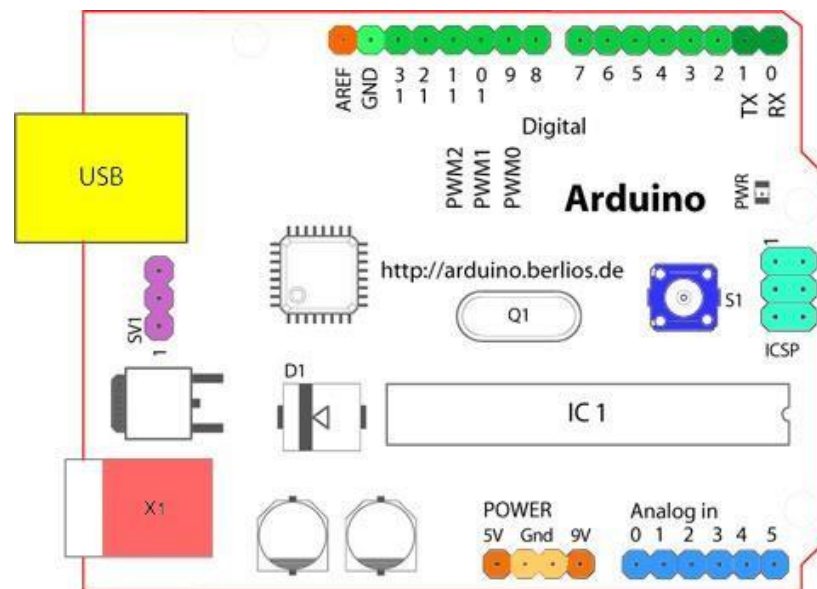


Fig 5.3 Structure of Arduino Board

Looking at the board from the top down, this is an outline of what you will see (parts of the board you might interact with in the course of normal use are highlighted)

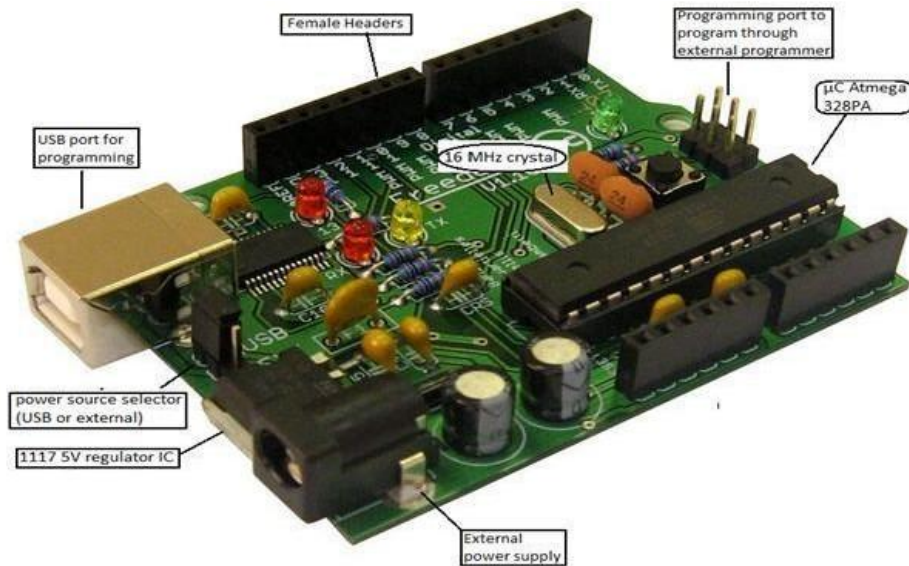


Fig 5.4 Arduino Board

Starting clockwise from the top centre:

- Analog Reference pin (orange)
- Digital Ground (light green)
- Digital Pins 2-13 (green)
- Digital Pins 0-1/Serial In/Out - TX/RX (dark green) - These pins cannot be used for digital i/o (Digital Read and Digital Write) if you are also using serial communication (e.g. Serial begin).
- Reset Button - S1 (dark blue)
- In-circuit Serial Programmer (blue-green)
- Analog In Pins 0-5 (light blue)
- Power and Ground Pins (power: orange, grounds: light orange)
- External Power Supply In (9-12VDC) - X1 (pink)
- Toggles External Power and USB Power (place jumper on two pins closest to desired supply) - SV1 (purple)
- USB (used for uploading sketches to the board and for serial communication between the board and the computer; can be used to power the board) (yellow)

DIGITAL PINS

In addition to the specific functions listed below, the digital pins on an Arduino board can be used for general purpose input and output via the pin Mode(), Digital Read(), and Digital Write() commands. Each pin has an internal pull-up resistor which can be turned on and off using digital Write () (w/ a value of HIGH or LOW, respectively) when the pin is configured as an input. The maximum current per pin is 40mA.

- **Serial: 0 (RX) and 1 (TX).** Used to receive (RX) and transmit (TX) TTL serial data. On the Arduino Diecimila, these pins are connected to the corresponding pins of the FTDI USB-to-TTL Serial chip. On the Arduino BT, they are connected to the corresponding pins of the WT11Bluetooth module. On the Arduino Mini and LilyPad Arduino, they are intended for use with an external TTL serial module (e.g. the Mini-USB Adapter).
- **External Interrupts: 2 and 3.** These pins can be configured to trigger an interrupt on a low value, a rising or falling edge, or a change in value. See the attach Interrupt () function for details.
- **PWM: 3, 5, 6, 9, 10, and 11** Provide 8-bit PWM output with the analog Write () function. On boards with an ATmega8, PWM output is available only on pins 9, 10, and 11.
- **BT Reset: 7.** (Arduino BT-only) Connected to the reset line of the blue tooth module.
- **SPI: 10 (SS), 11 (MOSI), 12 (MISO), 13 (SCK).** These pins support SPI communication, which, although provided by the underlying hardware, is not currently included in the Arduino language.
- **LED: 13.** On the Diecimila and LilyPad, there is a built-in LED connected to digital pin 13. When the pin is HIGH value, the LED is on, when the pin is LOW, it's off.

ANALOG PINS

In addition to the specific functions listed below, the analog input pins support 10-bit analog-to-digital conversion (ADC) using the analog Read () function. Most of the analog inputs can also be used as digital pins: analog input 0 as digital pin 14 through analog input 5 as digital pin 19. Analog inputs 6 and 7 (present on the Mini and BT) cannot be used as digital pins.

- **I²C: 4 (SDA) and 5 (SCL).** Support I²C (TWI) communication using the Wire library (documentation on the Wiring website).

POWER PINS

- **VIN** (sometimes labelled "9V"): The input voltage to the Arduino board when it's using an external power source (as opposed to 5 volts from the USB connection or other regulated power source). You can supply voltage through this pin, or, if supplying voltage via the power jack, access it through this pin. Also note that the Lily Pad has no VIN pin and accepts only a regulated input.
- **5V:** The regulated power supply used to power the microcontroller and other components on the board. This can come either from VIN via an on-board regulator, or be supplied by USB or another regulated 5V supply.
- **3V3** (Diecimila-only): A 3.3-volt supply generated by the on-board FTDI chip.
- **GND:** Ground pins.

OTHER PINS

- **AREF:** Reference voltage for the analog inputs. Used with analog Reference().
- **Reset:** (Diecimila-only) Bring this line LOW to reset the microcontroller. Typically used to add a reset button to shields which block the one on the board.

PIN DIAGRAM

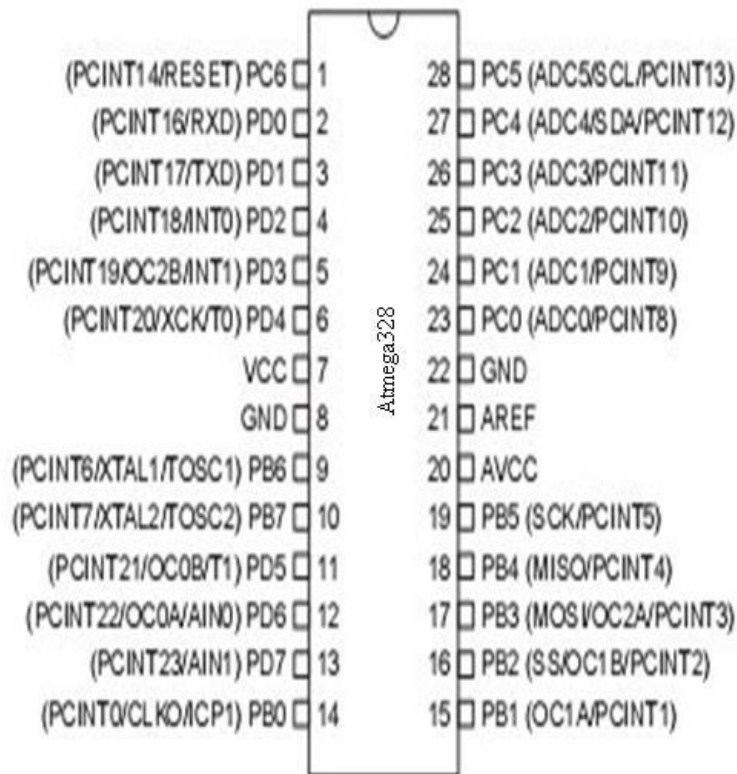


Fig 5.5 Pin Configuration of Atmega328

PIN DESCRIPTION

VCC:

Digital supply voltage

GND:

Ground

Port A (PA7-PA0):

Port A serves as the analog inputs to the A/D Converter. Port A also serves as an 8-bit bi-directional I/O port, if the A/D Converter is not used. Port pins can provide internal pull-up resistors (selected for each bit). The Port A output buffers have

symmetrical drive characteristics with both high sink and source capability. When pins PA0 to PA7 are used as inputs and are externally pulled low, they will source current if the internal pull-up resistors are activated. The Port A pins are tri-stated when a reset condition becomes active, even if the clock is not running.

Port B (PB7-PB0):

Port B is an 8-bit bi-directional I/O port with internal pull-up resistors (selected for each bit). The Port B output buffers have symmetrical drive characteristics with both high sink and source capability. As inputs, Port B pins that are externally pulled low will source current if the pull-up resistors are activated. The Port B pins are tri-stated when a reset condition becomes active, even if the clock is not running. Port B also serves the functions of various special features of the ATmega32.

Port C (PC7-PC0):

Port C is an 8-bit bi-directional I/O port with internal pull-up resistors (selected for each bit). The Port C output buffers have symmetrical drive characteristics with both high sink and source capability. As inputs, Port C pins that are externally pulled low will source current if the pull-up resistors are activated. The Port C pins are tri-stated when a reset condition becomes active, even if the clock is not running. If the JTAG interface is enabled, the pull-up resistors on pins PC5 (TDI), PC3 (TMS) and PC2 (TCK) will be activated even if a reset occurs. The TD0 pin is tri-stated unless TAP states that shift out data are entered. Port C also serves the functions of the JTAG interface.

Port D (PD7-PD0):

Port D is an 8-bit bi-directional I/O port with internal pull-up resistors (selected for each bit). The Port D output buffers have symmetrical drive characteristics with both high sink and source capability. As inputs, Port D pins that are externally pulled low will source current if the pull-up resistors are activated. The Port D pins

are tri-stated when a reset condition becomes active, even if the clock is not running. Port D also serves the functions of various special features of the ATmega32.

Reset (Reset Input):

A low level on this pin for longer than the minimum pulse length will generate a reset, even if the clock is not running. Shorter pulses are not guaranteed to generate a reset.

XTAL1:

Input to the inverting Oscillator amplifier and input to the internal clock operating circuit.

XTAL2:

Output from the inverting Oscillator amplifier.

AVCC:

AVCC is the supply voltage pin for Port A and the A/D Converter. It should be externally connected to VCC, even if the ADC is not used. If the ADC is used, it should be connected to VCC through a low-pass filter.

AREF:

AREF is the analog reference pin for the A/D Converter.

FEATURES

- 1.8-5.5V operating range
- Up to 20MHz
- Part: ATMEGA328P-AU
- 32kB Flash program memory
- 1kB EEPROM
- 2kB Internal SRAM

- 8-bit Timer/Counters
- 16-bit Timer/Counter
- RTC with separate oscillator
- 6 PWM Channels
- 8 Channel 10-bit ADC
- Serial USART
- Master/Slave SPI interface
- 2-wire (I2C) interface
- Watchdog timer
- Analog comparator
- 23 IO lines
- Data retention: 20 years at 85C/ 100 years at 25C
- Digital I/O Pins are 14 (out of which 6 provide PWM output)
- Analog Input Pins are 6.
- DC Current per I/O is 40 mA
- DC Current for 3.3V Pin is 50mA

AVR CPU Core

The AVR core combines a rich instruction set with 32 general purpose working registers. All the 32 registers are directly connected to the Arithmetic Logic Unit (ALU), allowing two independent registers to be accessed in one single instruction executed in one clock cycle. The resulting architecture is more code efficient while achieving throughputs up to ten times faster than conventional CISC microcontrollers.

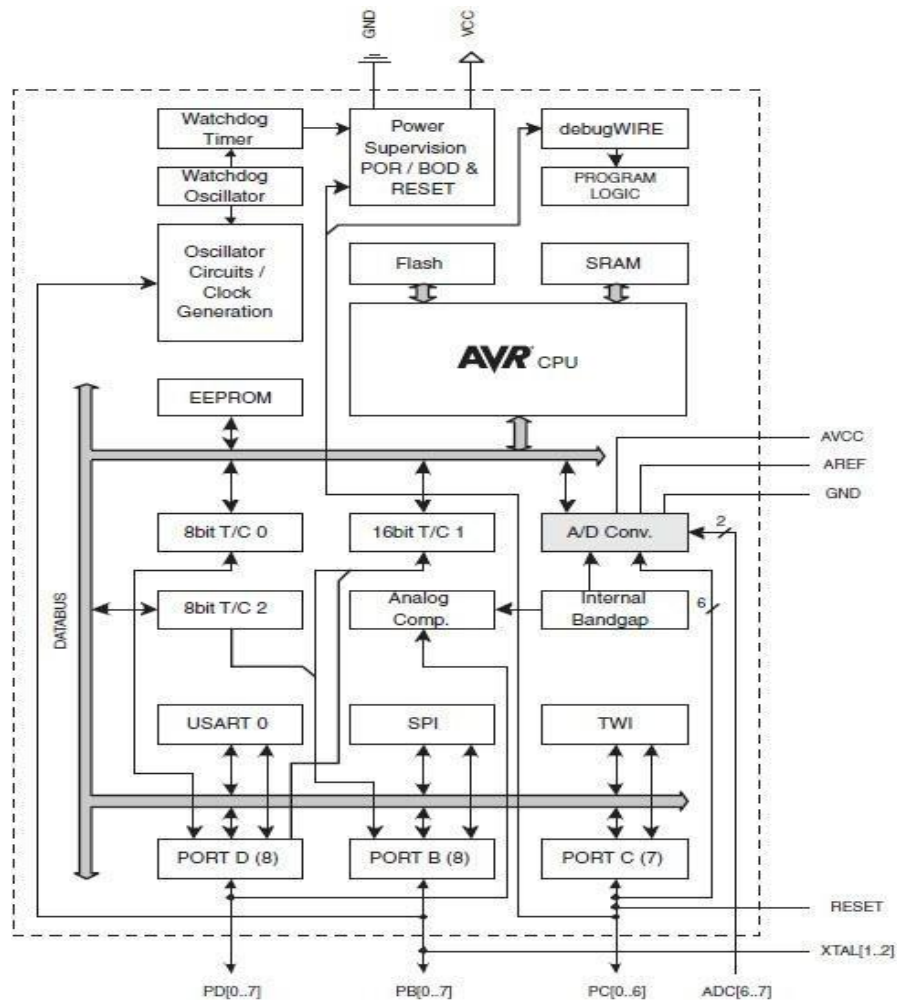


Fig 5.6 AVR CPU Core

OVERVIEW

This section discusses the AVR core architecture in general. The main function of the CPU core is to ensure correct program execution. The CPU must therefore be able to access memories, perform calculations, control peripherals, and handle interrupts.

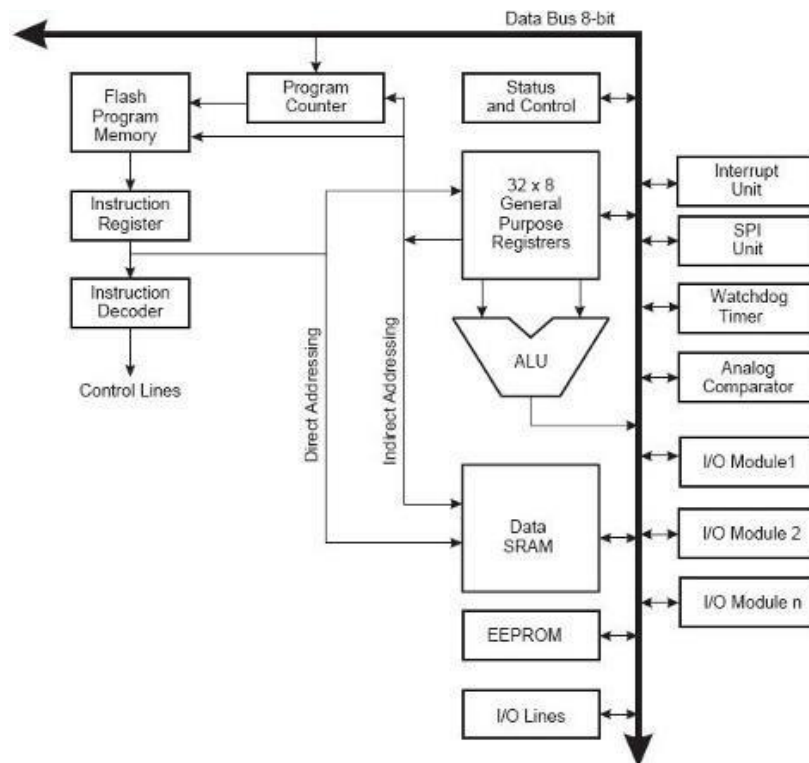


Fig 5.7 AVR core architecture

In order to maximize performance and parallelism, the AVR uses a Harvard architecture – with separate memories and buses for program and data. Instructions in the program memory are executed with a single level pipelining. While one instruction is being executed, the next instruction is pre-fetched from the program memory. This concept enables instructions to be executed in every clock cycle. The program memory is In-System Reprogrammable Flash memory. The fast-access Register File contains 32 x 8-bit general purpose working registers with a single clock cycle access time. This allows single-cycle Arithmetic Logic Unit (ALU) operation. In a typical ALU operation, two operands are output from the Register File, the operation is executed, and the result is stored back in the Register File– in one clock cycle.

Six of the 32 registers can be used as three 16-bit indirect address register pointers for Data Space addressing enabling efficient address calculations. One of these address pointers can also be used as an address pointer for look up tables in Flash program memory. These added function registers are the 16-bit X-, Y-, and Z-register,

described later in this section. The ALU supports arithmetic and logic operations between registers or between a constant and a register. Single register operations can also be executed in the ALU. After an arithmetic operation, the Status Register is updated to reflect information about the result of the operation. Program flow is provided by conditional and unconditional jump and call instructions, able to directly address the whole address space. Most AVR instructions have a single 16-bit word format. Every program memory address contains a 16- or 32-bit instruction. Program Flash memory space is divided in two sections, the Boot Program section and the Application Program section. Both sections have dedicated Lock bits for write and read/write protection. The SPM instruction that writes into the Application Flash memory section must reside in the Boot Program section. During interrupts and subroutine calls, the return address Program Counter (PC) is stored on the Stack. The Stack is effectively allocated in the general data SRAM, and consequently the Stack size is only limited by the total SRAM size and the usage of the SRAM. All user programs must initialize the SP in the Reset routine (before subroutines or interrupts are executed). The Stack Pointer (SP) is read/write accessible in the I/O space. The data SRAM can easily be accessed through the five different addressing modes supported in the AVR architecture.

The memory spaces in the AVR architecture are all linear and regular memory maps. A flexible interrupt module has its control registers in the I/O space with an additional Global Interrupt Enable bit in the Status Register. All interrupts have a separate Interrupt Vector in the Interrupt Vector table. The interrupts have priority in accordance with their Interrupt Vector position. The lower the Interrupt Vector address, the higher the priority.

ALU – Arithmetic Logic Unit

The high-performance AVR ALU operates in direct connection with all the 32 general purpose working registers. Within a single clock cycle, arithmetic operations between general purpose registers or between a register and an immediate are executed. The ALU operations are divided into three main categories – arithmetic, logical, and bit functions. Some implementations of the architecture also provide a

powerful multiplier supporting both Signed/unsigned multiplication and fractional format. See the “Instruction Set” section for a detailed description.

STATUS REGISTER

The Status Register contains information about the result of the most recently executed arithmetic instruction. This information can be used for altering program flow in order to perform conditional operations. Note that the Status Register is updated after all ALU operations, as specified in the Instruction Set Reference. This will in many cases remove the need for using the dedicated compare instructions, resulting in faster and more compact code. The Status Register is not automatically stored when entering an interrupt routine and restored when returning from an interrupt. This must be handled by software.

The AVR Status Register – SREG is defined as:

Bit	7	6	5	4	3	2	1	0	
	I	T	H	S	V	N	Z	C	SREG
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

Fig AVR Status Register

Bit 7 – I: Global Interrupt Enable

The Global Interrupt Enable bit must be set for the interrupts to be enabled. The individual interrupt enable control The Global Interrupt Enable bit must be set for the interrupts to be enabled. The individual interrupt enable control is then performed in separate control registers. If the Global Interrupt Enable Register is cleared, none of the interrupts are enabled independent of the individual interrupt enable settings. The I-bit is cleared by hardware after an interrupt has occurred, and is set by the RETI instruction to enable subsequent interrupts. The I-bit can also be set and cleared by the application with the SEI and CLI instructions, as described in the instruction set reference.

Bit 6 – T: Bit Copy Storage

The Bit Copy instructions BLD (Bit Load) and BST (Bit Store) use the T-bit as source or destination for the operated bit. A bit from a register in the Register File can be copied into T by the BST instruction, and a bit in T can be copied into a bit in a register in the Register File by the BLD instruction.

Bit 5 – H: Half Carry Flag

The Half Carry Flag H indicates a Half Carry in some arithmetic operations. The Half Carry Flag H indicates a Half Carry in some arithmetic operations. Half Carry is useful in BCD arithmetic. See the “Instruction Set Description” for detailed information.

Bit 4 – S: Sign Bit, $S = N \oplus V$

The S-bit is always an exclusive or between the Negative Flag N and the Two’s Complement Overflow Flag V. See the “Instruction Set Description” for detailed information.

Bit 3 – V: Two’s Complement Overflow Flag

The Two’s Complement Overflow Flag V supports two’s complement arithmetic.

Bit 2 – N: Negative Flag

The Negative Flag N indicates a negative result in an arithmetic or logic operation.

Bit 1 – Z: Zero Flag

The Zero Flag Z indicates a zero result in an arithmetic or logic operation.

Bit 0 – C: Carry Flag

The Carry Flag C indicates a carry in an arithmetic or logic operation.

GENERAL PURPOSE REGISTER FILE

The Register File is optimized for the AVR Enhanced RISC instruction set. In order to achieve the required performance and flexibility, the following input/output schemes are supported by the Register File:

- One 8-bit output operand and one 8-bit result input
- Two 8-bit output operands and one 8-bit result input
- Two 8-bit output operands and one 16-bit result input
- One 16-bit output operand and one 16-bit result input

Most of the instructions operating on the Register File have direct access to all registers, and most of them are single cycle instructions. As shown in Figure 6-2, each register is also assigned a data memory address, mapping them directly into the first 32 locations of the user data space. Although not being physically implemented as SRAM locations, this memory organization provides great flexibility in excess of the registers, as the X-, Y-, and Z-pointer registers can be set to index any register.

		7	0	Addr.	
General Purpose Working Registers		R0		0x00	
		R1		0x01	
		R2		0x02	
		...			
		R13		0x0D	
		R14		0x0E	
		R15		0x0F	
		R16		0x10	
		R17		0x11	
		...			
		R26		0x1A	X-register Low Byte
		R27		0x1B	X-register High Byte
		R28		0x1C	Y-register Low Byte
		R29		0x1D	Y-register High Byte
		R30		0x1E	Z-register Low Byte
		R31		0x1F	Z-register High Byte

X-, Y-, and Z-register

The registers R26:R31 have some added functions to their general-purpose usage. These registers are 16-bit address pointers for indirect addressing of the data space. The three indirect address registers X, Y, and Z are defined as described in

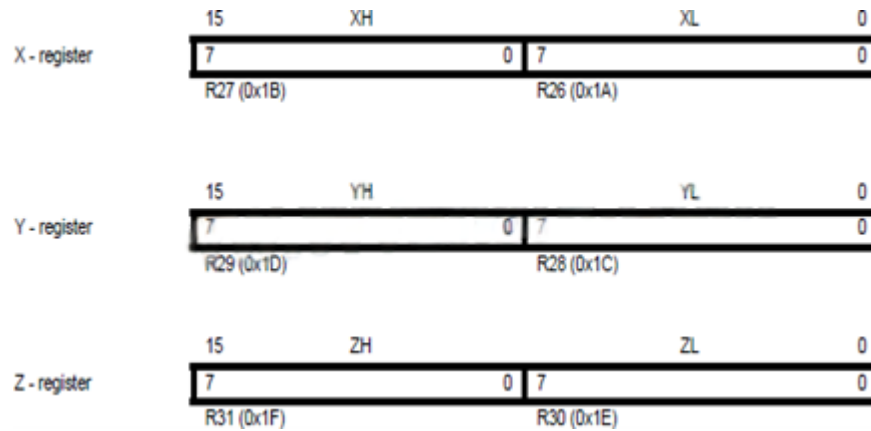


Fig X-, Y-, and Z-Registers

In the different addressing modes these address registers have functions as fixed displacement, automatic increment, and automatic decrement.

Stack Pointer

The Stack is mainly used for storing temporary data, for storing local variables and for storing return addresses after interrupts and subroutine calls. Note that the Stack is implemented as growing from higher to lower memory locations. The Stack Pointer Register always points to the top of the Stack. The Stack Pointer points to the data SRAM Stack area where the Subroutine and Interrupt Stacks are located. A Stack PUSH command will decrease the Stack Pointer.

The Stack in the data SRAM must be defined by the program before any subroutine calls are executed or interrupts are enabled. Initial Stack Pointer value equals the last address of the internal SRAM and the Stack Pointer must be set to point above start of the SRAM.

Instruction	Stack pointer	Description
PUSH	Decrement by 1	Data is pushed onto the stack
CALL ICALL RCALL	Decrement by 2	Return address is pushed onto the stack with a subroutine call or interrupt
POP	Increment by 1	Data is popped from the stack
RET RETI	Increment by 2	Return address is popped from the stack with return from subroutine or return from interrupt

Fig 5.1 Table Stack Pointer instructions

The AVR ATmega128A Stack Pointer is implemented as two 8-bit registers in the I/O space. The number of bits actually used is implementation dependent. Note that the data space in some implementations of the AVR architecture is so small that only SPL is needed. In this case, the SPH Register will not be present and SPL - Stack Pointer High and Low Register.

Bit	15	14	13	12	11	10	9	8	
	SP15	SP14	SP13	SP12	SP11	SP10	SP9	SP8	SPH
	SP7	SP6	SP5	SP4	SP3	SP2	SP1	SP0	SPL
	7	6	5	4	3	2	1	0	
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	
	0	0	0	0	0	0	0	0	

Fig 5.2 SPH and SPL - Stack Pointer High and Low Register

INTERRUPT RESPONSE TIME

The interrupt execution response for all the enabled AVR interrupts is four clock cycles minimum. After four clock cycles the program vector address for the actual interrupt handling routine is executed. During this four clock cycle period, the Program Counter is pushed onto the Stack. The vector is normally a jump to the interrupt routine, and this jump takes three clock cycles. If an interrupt occurs during execution of a multi-cycle instruction, this instruction is completed before the interrupt is served.

If an interrupt occurs when the MCU is in sleep mode, the interrupt execution response time is increased by four clock cycles. This increase comes in addition to the start-up time from the selected sleep mode.

AVR Memories

This section describes the different memories in the ATmega328. The AVR architecture has two main memory spaces, the Data Memory and the Program Memory space. In addition, the ATmega328 features an EEPROM Memory for data storage. All three memory spaces are linear and regular.

In-System Reprogrammable Flash Program Memory:

The ATmega328 contains 4/8/16/32Kbytes On-chip In-System Reprogrammable Flash memory for program storage. Since all AVR instructions are 16 or 32 bits wide, the Flash is organized as 2/4/8/16K x 16. For software security, the Flash Program memory space is divided into two sections, Boot Loader Section and Application Program Section. The Flash memory has an endurance of at least 10,000 write/erase cycles. The ATmega328 Program Counter (PC) is 11/12/13/14 bits wide, thus addressing the 2/4/8/16K program memory locations.

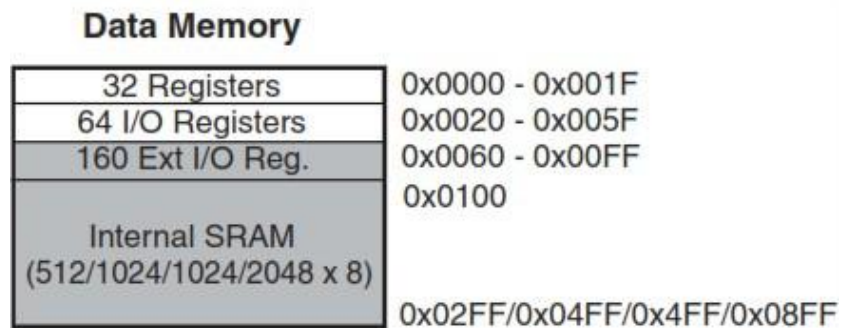
SRAM Data Memory:

ATmega328 is a complex microcontroller with more peripheral units than can be supported within the 64 locations reserved in the Opcode for the IN and OUT instructions. For the Extended I/O space from 0x60 - 0xFF in SRAM, only the ST/STS/STD and LD/LDS/LDD instructions can be used.

The lower 768/1280/1280/2303 data memory locations address both the Register File, the I/O memory, Extended I/O memory, and the internal data SRAM. The first 32 locations address the Register File, the next 64 location the standard I/O memory, then 160 locations of Extended I/O memory, and the next 512/1024/1024/2048 locations address the internal data SRAM. The five different addressing modes for the data memory cover: Direct, Indirect with Displacement, Indirect, Indirect with Pre-decrement, and Indirect with Post-increment. In The

Register File, Registers R26 to R31 Feature the indirect addressing pointer registers. The direct addressing reaches the entire data space. The Indirect with Displacement mode reaches 63 address locations from the base address given by the Y- or Z register.

When using registers indirect addressing modes with automatic pre-decrement and post-increment, the address registers X, Y, and Z are decremented or incremented. The 32 general purpose working registers, 64 I/O Registers, 160 Extended I/O Registers, and the 512/1024/1024/2048 bytes of internal data SRAM in the ATmega328 are all accessible through all these addressing modes.



Data Memory Map

5.3 ESP32 CAM

The ESP32-CAM is a development board with an ESP32-S chip, an OV2640 camera, micro SD card slot and several GPIOs to connect peripherals. It allows you to set up a video streaming web server, build a surveillance camera, take photos, face recognition and detection, and much more.

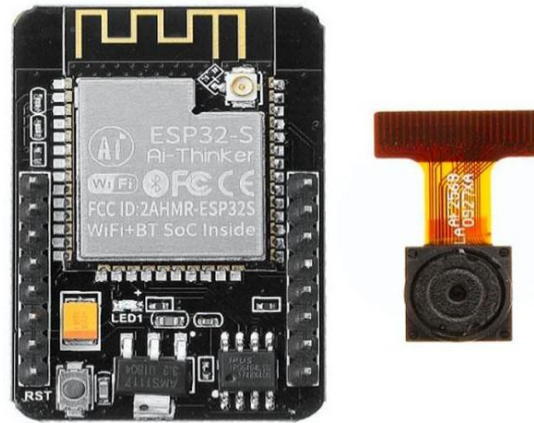


Fig 5.8 ESP32 Cam

FEATURES

- Product Name: ESP32-CAM. WiFi+Bluetooth module: ESP-32S.
- Camera Module: OV2640 2MP.
- Flash Light: LED Built-in on Board.
- Operating Voltage: 3.3/5 Vdc.
- Onboard TF card slot, supports up to 4G TF card for data storage .
- RAM: Internal 512KB + External 4MB PSRAM.
- Power consumption:
 - Flash off: 180mA@5V.
 - Flash on and brightness max: 310mA@5V.
 - Deep-Sleep: as low as 6mA@5V.
 - Modern-Sleep: as low as 20mA@5V.
 - Light-Sleep: as low as 6.7mA@5V
- Dimensions: 40.5mm x 27mm x 4.5mm.

5.4 ULTRASONIC SENSOR

Ultrasonic is an excellent way of figuring out what's in the immediate vicinity of your Arduino. The basics of using ultrasound are like this: you shoot out a sound, wait to hear it echo back, and if you have your timing right, you'll know if anything is out there and how far away it is. This is called echolocation and it's how bats and dolphins find objects in the dark and underwater, though they use lower frequencies than you can use with your Arduino.



Fig 5.9 Ultrasonic Sensor

PRINCIPLE

Ultrasonic sensors can measure distance and detect the presence of an object without making physical contact. They do so by producing and monitoring an ultrasonic echo. Depending on the sensor and object properties, the effective range in air is between a few centimeters up to several meters. The ultrasonic sensor (or transducer) generates and emits ultrasonic pulses that are reflected back towards the sensor by an object that is within the field of view of the sensor.

5.5 SOIL MOISTURE SENSOR

The soil moisture sensor or the hygrometer is usually used to detect the humidity of the soil. So, it is perfect to build an automatic watering system or to monitor the soil moisture of your plants. The sensor is set up by two pieces: the electronic board and the sensor probe with two pads, that detects the water content.

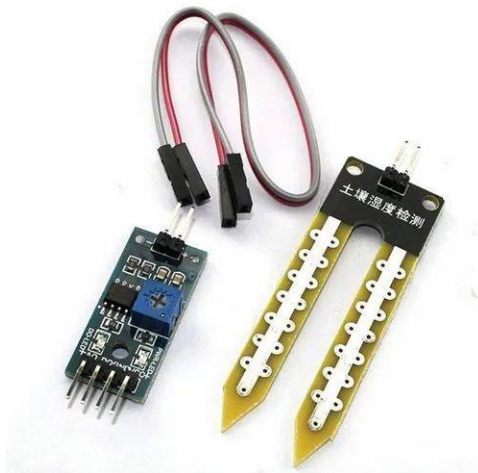


Fig 5.10 Soil Moisture Sensor

Description

- Interface: 4-Wire.
- Operating Voltage: 3.3V~5Vdc.
- Output: Digital and Analogue.
- Cable Length: ~21cm.
- Panel PCB Dimension: 3cm x 1.5cm.
- Soil Probe Dimension: 6cm x 3cm

FEATURES

- High accuracy sensor
- Fast response sensor
- Simple soil insertion (vertical and Horizontal orientation)

- Programmable start time and data
- Miniature Size
- Weather proof enclosure
- User calibration through MadgeTech Software
- Probe is powered from data recorder's battery. No external power required.

5.6 SMTP

SMTP (Simple Mail Transfer Protocol)

SMTP is a standard electronic-mail protocol that handles the sending of mail from one SMTP to another SMTP server. To accomplish the transport, the SMTP server has its own MX (mail exchanger) record in the DNS database that corresponds to the domain for which it is configured to receive mail.

When equipped for two-way communication, mail clients are configured with the address of a POP3 server to receive mail and the address of an SMTP server to send mail.

SMTP uses TCP for communication and operates on port 25. Simple Mail Transfer Protocol (SMTP) is the application-layer protocol used for transmitting e-mail messages. SMTP is capable of receiving e-mail messages, but it's limited in its capabilities. The most common implementations of SMTP are in conjunction with either POP3 or IMAP4. For example, users download an e-mail message from a POP3 server, and then transmit messages via an SMTP server

5.7 WI-FI MODULE

ESP-01 Wi-Fi module is developed by Ai-thinker Team. Core processor ESP8266 in smaller sizes of the module encapsulates Tensilica L106 integrates industry-leading

ultra-low power 32-bit MCU micro, with the 16-bit short mode, Clock speed support 80 MHz, 160 MHz, supports the RTOS, integrated Wi-Fi MAC/BB/RF/PA/LNA, on-board antenna.

The module supports standard IEEE802.11 b/g/n agreement, complete TCP/IP protocol stack. Users can use the add modules to an existing device networking, or building a separate network controller.

ESP8266 is high integration wireless SOCs, designed for space and power constrained mobile platform designer. It provides unsurpassed ability to embed Wi-Fi capabilities within other systems, or to function as a standalone application, with the lowest cost, and minimal space requirement.

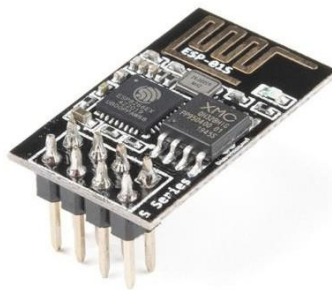


Fig 5.11 Wi-Fi Module

FEATURES

- 802.11 b/g/n
- Integrated low power 32-bit MCU
- Integrated 10-bit ADC
- Integrated TCP/IP protocol stack
- Integrated TR switch, balun, LNA, power amplifier and matching network
- Integrated PLL, regulators, and power management units
- Supports antenna diversity
- Wi-Fi 2.4 GHz, support WPA/WPA2

- Support STA/AP/STA+AP operation modes
- Support Smart Link Function for both Android and iOS devices
- Support Smart Link Function for both Android and iOS devices
- SDIO 2.0, (H) SPI, UART, I2C, I2S, IRDA, PWM, GPIO
- STBC, 1x1 MIMO, 2x1 MIMO
- A-MPDU & A-MSDU aggregation and 0.4s guard interval
- Deep sleep power < 5Ua
- Wake up and transmit packets in < 2ms
- Standby power consumption of < 1.0mW (DTIM3)
- +20dBm output power in 802.11b mode.
- Operating temperature range -40C ~ 125C

5.8 LCD (LIQUID CRYSTAL DEVICE):

LCD (liquid crystal display) is the technology used for displays in notebook and other smaller computers. Like light-emitting diode (LED) and gas-plasma technologies, LCDs allow displays to be much thinner than cathode ray tube (CRT) technology. LCDs consume much less power than LED and gas-display displays because they work on the principle of blocking light rather than emitting it.



Fig 5.12 LCD

FEATURES

- E-blocks compatible
- Low cost
- Compatible with most I/O ports in the E-Block range (requires 5 I/O lines via 9-way D-type connector)
- Ease to develop programming code using Flow code icons

Fundamentals of Liquid Crystal Displays

The term liquid crystal is used to describe a substance in a state between liquid and solid but which exhibits the properties of both. Molecules in liquid crystals tend to arrange themselves until they all point in the same specific direction. This arrangement of molecules enables the medium to flow as a liquid. Depending on the temperature and particular nature of a substance, liquid crystals can exist in one of several distinct phases. Liquid crystals in a nematic phase, in which there is no spatial ordering of the molecules, for example, are used in LCD technology.

One important feature of liquid crystals is the fact that an electrical current affects them. A particular sort of nematic liquid crystal, called twisted nematics (TN), is naturally twisted. Applying an electric current to these liquid crystals will untwist them to varying degrees, depending on the current's voltage. LCDs use these liquid crystals because they react predictably to electric current in such a way as to control the passage of light.

The working of a simple LCD is shown in Figure 1. It has a mirror (A) in back, which makes it reflective. There is a piece of glass (B) with a polarizing film on the bottom side, and a common electrode plane (C) made of indium-tin oxide on top. A common electrode plane covers the entire area of the LCD. Above that is the layer of liquid crystal substance (D). Next comes another piece of glass (E) with an electrode in the shape of the rectangle on the bottom and, on top, another polarizing film (F), at a right angle to the first one.

5.9 BUZZER

Buzzer is usually like an alarm. Whenever we press the switch button it gives an output like an alarm sound and then activates the machine. Buzzer contains of two pins. The negative end is connected to the data pin of microcontroller. The positive end is connected to the Vcc in the microcontroller.



Fig 5.13 Buzzer

SOFTWARE IMPLEMENTATION

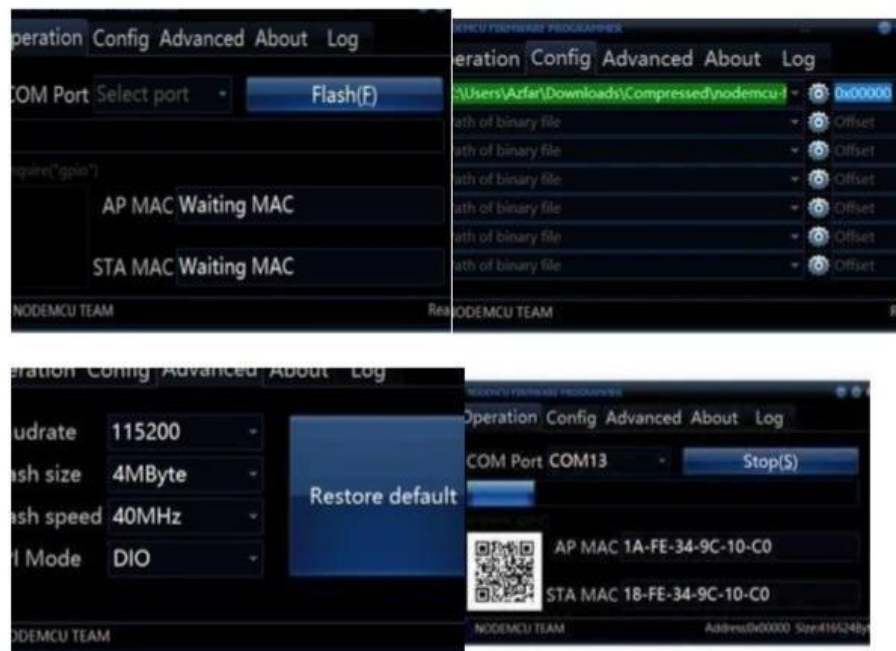
6.1 ARDUINO SOFTWARE

PROGRAMMING

Step1: Installing the Firmware

Programming

Step 1: Installing the Firmware



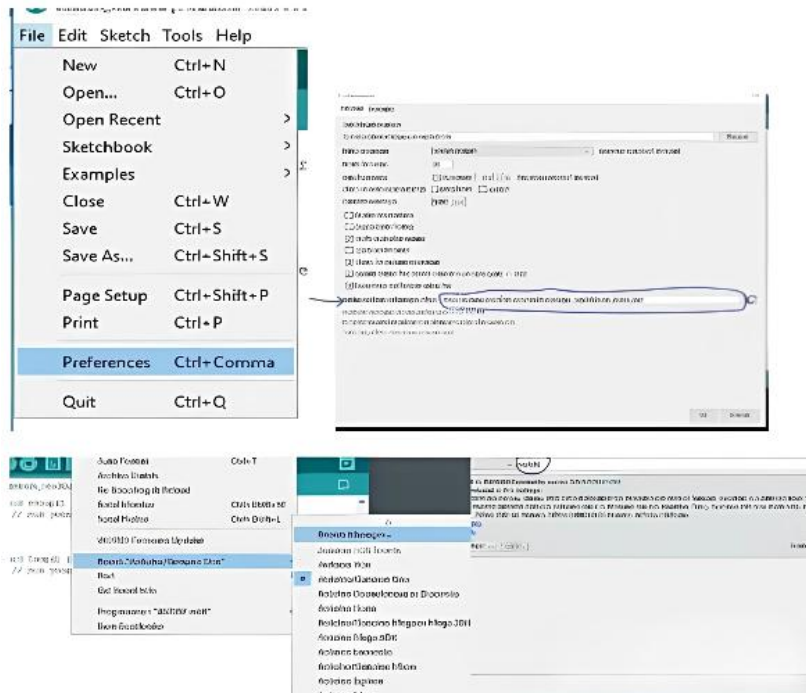
In Node MCU Boards the first thing you need is to install the Firmware to the board the following method works for all Node MCU Boards

1. Open the Node MCU flasher master folder than open the win32/win64 folder as your computer. Now open the folder Release than double click ESP8266Flasher.
2. Select the Comport.
3. Go to configure tab.
4. Click on the small gear and open up the firmware which you have

downloaded.

5. Go to the advanced tab and select the desired Baud rate.
6. Go to the Operation tab and click on Flash Button. Add Tip Ask Question Comment Download

Step2: Preparing the Arduino IDE



After installing the firmware, you are ready to do the programming with the ESP8266

1. Install the Arduino IDE
2. Open the Arduino IDE from the desktop icon
3. Click on File tab and then open preferences
4. In the additional Boards Manager URLs add the following link (http://arduino.esp8266.com/stable/package_esp8266com_index.json) and click OK

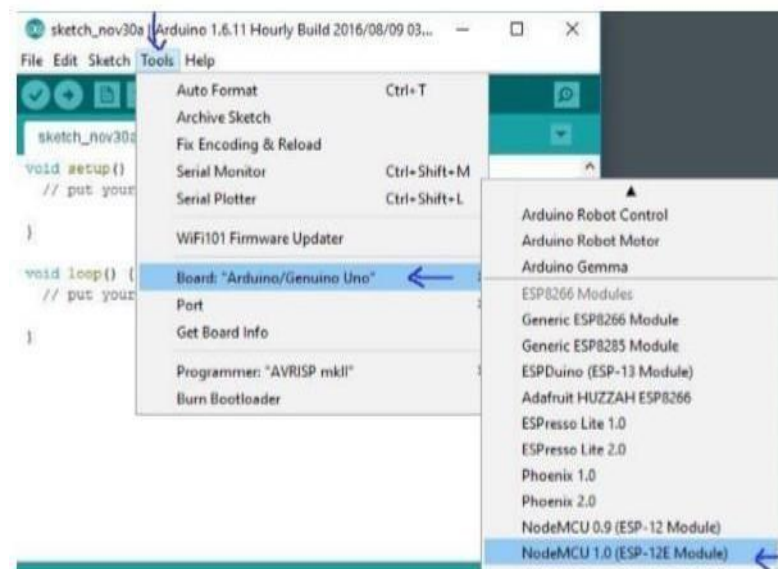
1. GotoTools>Boards>BoardsManager

2. Inthe search field type esp8266 click the esp8266 byESP8266 Community option and click Install

Step3: Code...

Now we can do whatever you want with your Node MCU board Following is an example for led blinking with Node MCU board via web server

- In arduino IDE go to tools>Boards>select NODE MCU 1.0 (ESP- 12E Module)
- Again, go to tools and select port.
- Change the Wi-Fi name and password from the following code.
- Now click on Upload button to upload the following code.
- Connect the led's positive leg on D9 pin of board and negative to the ground of the code.
- Power up the board and open the serial monitor from arduino IDE
- After connecting to the Wi-Fi, it will show you the IP address.
- Type that IP address on the web browser (Edge, Chrome, Firefox etc.,)
- A webpage will open you can change the status of LED by turning it ON or OFF.



COPY, PASTE AND UPLOAD THE TUTORIAL SKETCH

The sketch is one that comes as an example from ESP8266.COM.

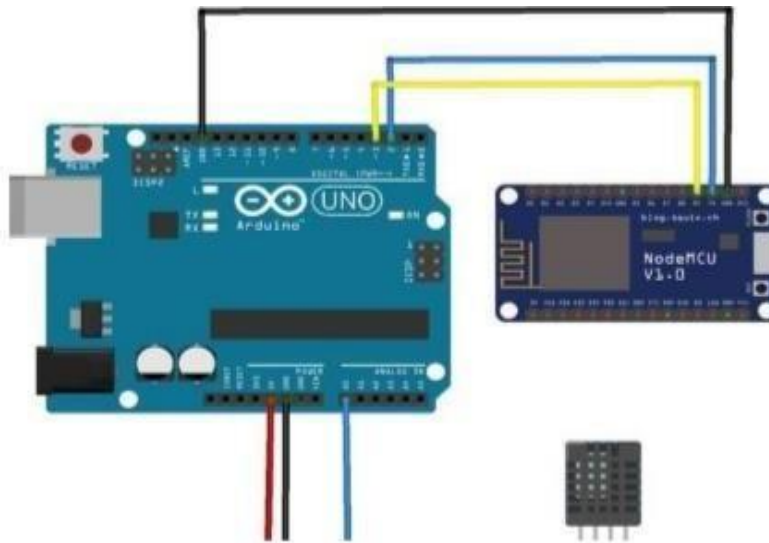
```
#include "ESP8266WiFi.h"

void setup() {
    Serial.begin(115200);
    // Set WiFi to station mode and disconnect from an AP if it was
    previously connected
    WiFi.mode(WIFI_STA);
    WiFi.disconnect();
    delay(2000);
    Serial.println("Setup done");
}

void loop() {
    Serial.println("scan start");

    int n = WiFi.scanNetworks(); // WiFi.scanNetworks will return the
    number of networks found
    Serial.println("scan done");
    if (n == 0)
        Serial.println("no networks found");
    else
    {
        Serial.print(n);
        Serial.println(" networks found");
        for (int i = 0; i < n; ++i)
        {
            // Print SSID and RSSI for each network found
        }
    }
}
```

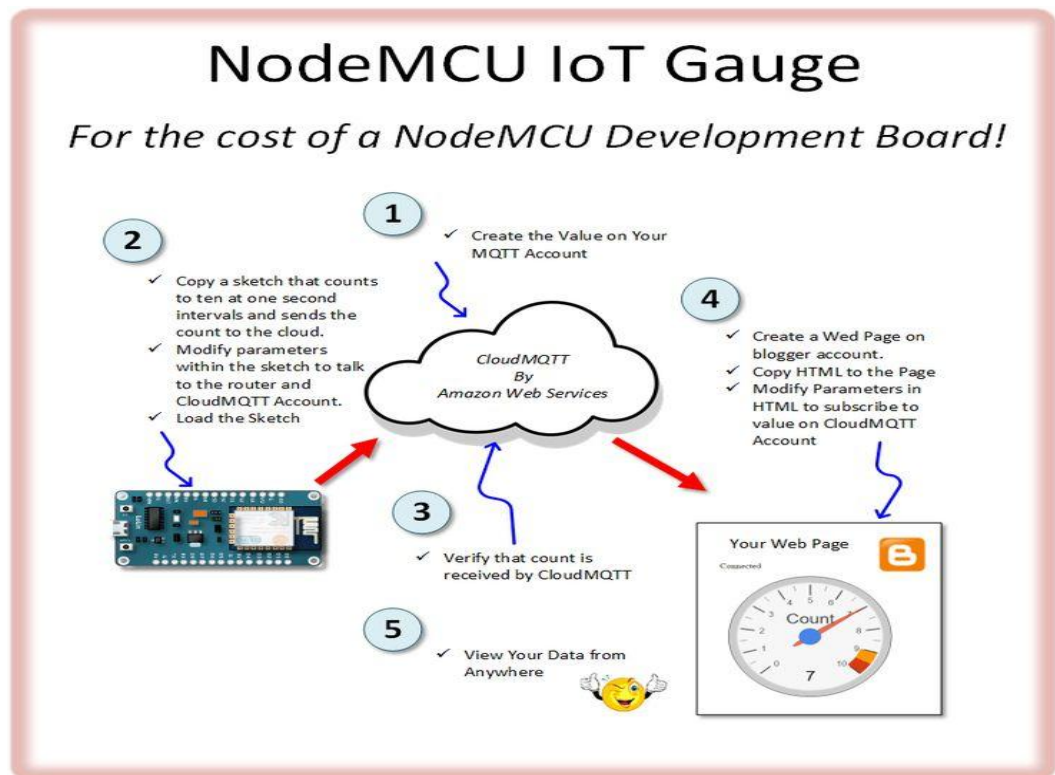
```
Serial.print(i + 1);  
Serial.print(": ");  
Serial.print(WiFi.SSID(i));  
Serial.print(" ");  
Serial.print(WiFi.RSSI(i));  
Serial.print(" ");  
Serial.println((WiFi.encryptionType(i) == ENC_TYPE_NONE)? " ":  
"");  
    delay(10);  
}  
Serial.println("");  
  
// Wait a bit before scanning again  
delay(5000);  
}
```



Serial Communication between Node MCU and Arduino

PROJECTS AND APPLICATIONS

Node MCU V3 is mainly used in the Wi-Fi Applications which most of the other embedded modules fail to process unless incorporated with some external Wi-Fi protocol. Following are some major applications used for Node MCU V3.



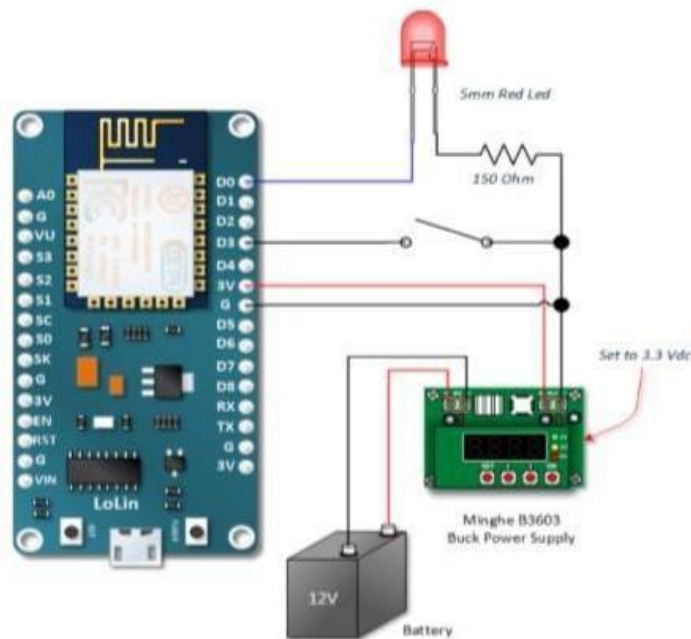
- Internet Smoked Alarm
- VR Tracker
- Octopod
- Serial Port Monitor
- ESP Lamp
- Incubator Controller
- IoT home automation
- Security Alarms

Application Example:

Node MCU ESP-12E Arduino IDE Digital Input Tutorial

CONNECT THE CIRCUIT

You may wish to power your Node MCU another way. You can read about it [HERE](#).



COPY, PASTE AND UPLOAD THE CODE

The code is real simple works as follows:

The input is read from switch pin. If switch is closed, it will read a low.

The LED output pin is set to the opposite of the switch pin. If the switch is closed, the output pin will be set to a high. A high will turn on the LED.

```
int switchPin = 0;
int LedPin = 16;
int switchValue;

void setup() {
  pinMode(LedPin, OUTPUT);
  pinMode(switchPin, INPUT_PULLUP);
}

void loop() {
  // Read the switch value
  switchValue = digitalRead(switchPin);
  /* Set the LED output pin the opposit of what is read on the sw
  itch
  * input pin
  */
  digitalWrite(LedPin, !switchValue);
}
```

6.2 PROTEUS

INTRODUCTION:

Generally, we are listening the words PCB's, PCB layout, PCB designing, etc. But what is PCB? Why we are using this PCB? We want to know about all these things as an electronic engineer. PCB means Printed Circuit Board. This is a circuit board with printed copper layout connections. These PCBs are two types. One is dotted PCB and another one is layout PCB. The two examples are shown in below.

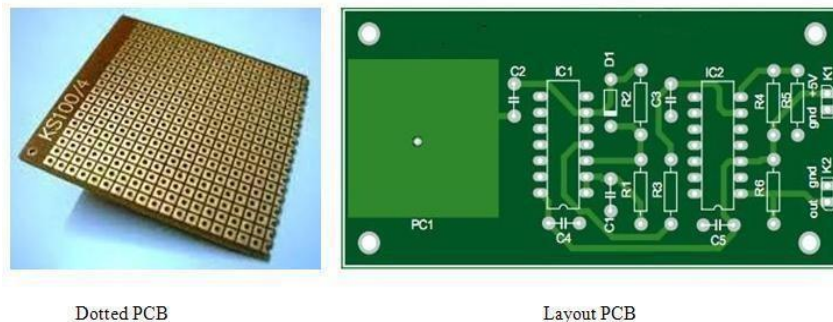


Fig 6.1 Dotted PCB and Layout PCB

What is the main difference between the dotted PCB and layout PCB?

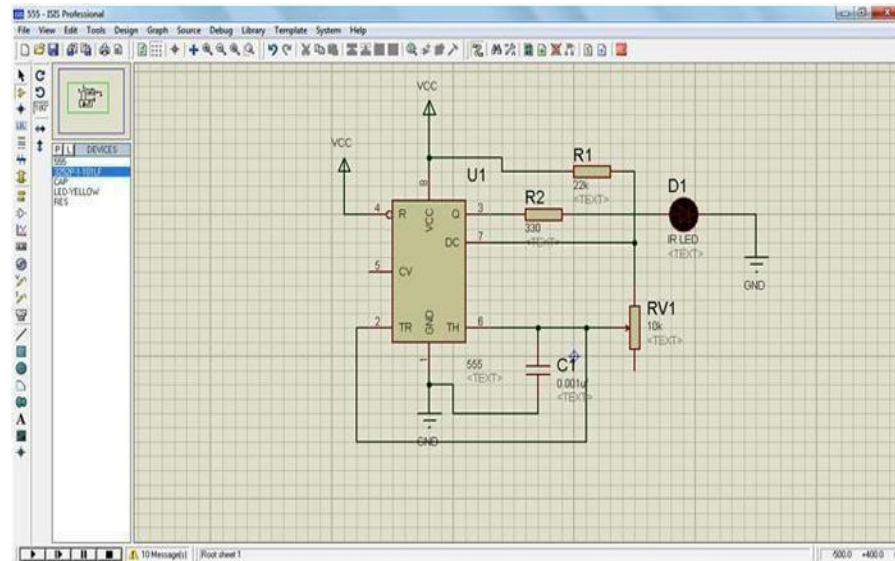
In dotted PCB board only, dots are available. According to our requirement we can place or insert the components in those holes and attach the components with wires and soldering lid. In this dotted PCB we can make the circuit as out wish but it is very hard to design. There are so many difficulties are there. Those are connecting the proper pins, avoiding shot connections and etc. Coming to the layout PCB this is simple to design. First, we select the our circuit and by using different PCB designing software's, design the layout of the circuit and by itching process preparing the copper layout of our circuit and solder the components in the correct places. It is simple to design, take less time to design, no shortages, looking nice and perfect.

Up to now we have discussed about types of PCB's and difference between the types. Now we can discuss about PCB designing software. There are so many PCB designing software's available. Some are Express PCB, eagle PCB, PCB Elegance, free PCB, open circuit design, zenith PCB and Proteus etc. Apart from remaining Proteus is different. Proteus is design suit and PCB layout designing software. In Proteus we can design any circuit and simulate the circuit and make PCB layout for that circuit.

Introduction to Proteus:

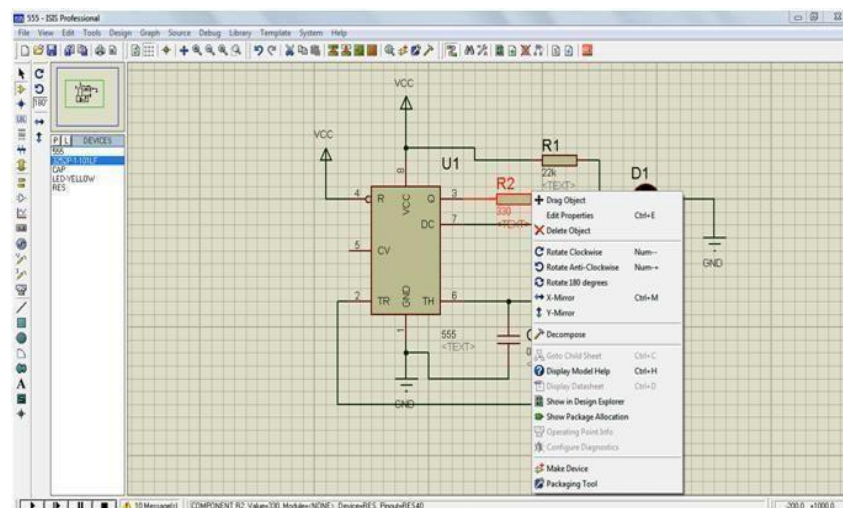
Proteus professional is a software combination of ISIS schematic capture program and ARES PCB layout program. This is a powerful and integrated development environment. Tools in this suit are very easy to use and these tools are very useful in education and professional PCBd esigning.

As professional PCB designing software with integrated space-based auto router, it the curser at the component pin end then draw the connections with that pen symbol. Connect all the components according to circuit then that designed circuit is show in below image.



If any modifications want to do to the component place the mouse point and click on right button then option window will open. That is shown in below figure.

After completion of designing save with some mane and debug it. This is virtual simulation means without making circuit we can see the result in virtually through this software and we can **design the PCB layout** to our required circuit with this software.



RESULT

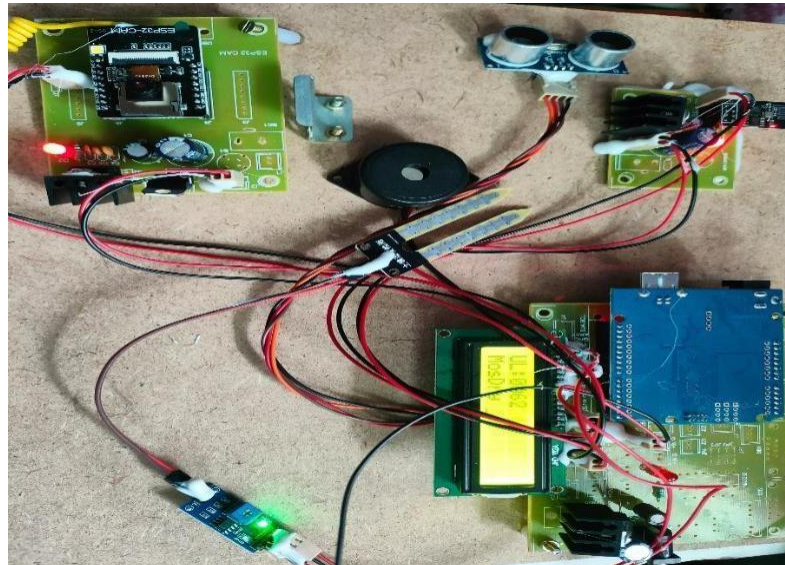


FIG 7.1 HARDWARE CONNECTION OF THE KIT

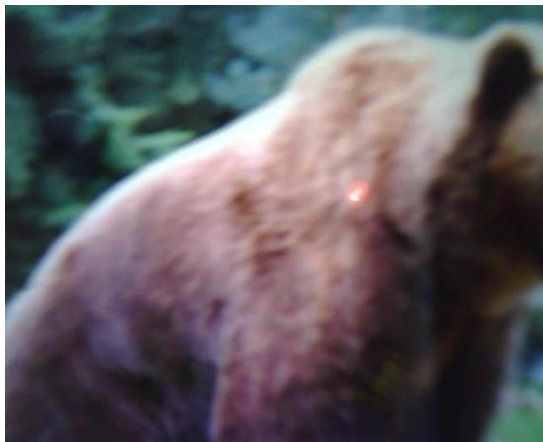


FIG7.2 OUTPUT IMAGE

```
Connecting to 192.168.4.1 port 23, please wait...  
Intruder U:00001  
Intruder U:00004  
Intruder U:00001  
Intruder U:00007  
Mos Wet  
Intruder U:00003  
Mos Wet  
Mos Wet  
Mos Wet  
Mos Wet  
Mos Wet
```

FIG7.3 OUTPUT VALUES

7.1 OPERATION

The operation of the kit follows as below the hardware kit consists of following components such as lcd, RPS, ESP-32 Cam, Soil Moisture sensor, ultrasonic sensor, buzzer, SMTP, Arduino, Wi-Fi Module firstly we have to download the mobile telnet app through it we will connect the kit after connecting through wi-fi (org) we have to open telnet app and connect through Internet Protocol address that belong to kit after that we have to place an object near to the ultra sonic sensor,the buzzer makes a sound when object near to the sensor the ESP-32 cam captures image and then send to mail through wifi module.

ADVANTAGES AND APPLICATIONS

8.1 ADVANTAGES

- 1. 24/7 Monitoring of Fields:** Uses sensors and cameras to monitor crop fields day and night
- 2. Immediate Detection and Alerts:** Sends real-time notifications to farmers when animals are detected.
- 3. Non-lethal and Eco-friendly:** Uses light, sound, or other harmless deterrents instead of traps or poisons.
- 4. Reduced Crop Damage:** Prevents animal intrusions before they cause significant damage.
- 5. Saves Labor and Time:** Minimizes the need for farmers to patrol fields manually, especially at night.
- 6. Cost-effective in the Long Run:** Initial setup cost is offset by the savings from reduced crop loss.
- 7. Data Collection and Analysis:** Can log animal activity patterns, helping to predict future threats.

8.2 APPLICATIONS

- Agricultural fields near forests or wildlife zones
- Remote farms
- Large-scale commercial farms
- NGO wildlife conflict zones
- Eco tourism farms

CONCLUSION AND FUTURE SCOPE

Using sensor networks and machine learning, our system offers a full smart agricultural system which monitors and manages crops. The system is made up of Internet of Things devices that monitor several environmental factors and a mobile application that enables users to control irrigation system, sensors in real time also helps the farmer managing natural disasters like agricultural fires. Additionally, the system uses machine learning models that suggest the best crops and fertilizers according to soil constraints and some other environmental factors. Through the integration of Internet of Things and machine learning, this system may assist farmers to make their decisions effectively, save their time, prevent accidents and overcome the traditional way of farming, also helps the farmer to avoid infections from fertilizer, improve crop yields, while reducing environmental effect and resource use.

These IoT and ML based recommendations will surely help the farmer understand more about the various factors, which help them in minimizing their costs, and make strategic decisions. This leads to a scalable and reliable solution, which will affect millions of people in India.

REFERENCE

1. G Sushanth, S Sujatha “IOT Based Smart Agriculture System”, 2018 International Conference on Wireless Communications, Signal Processing and Networking (Wi-SPNET).
2. Rahul Dagar, Subhranil Som, Sunil Kumar Khatri “Smart Farming – IoT in Agriculture”, 2018 International Conference on Inventive Research in Computing Applications (ICIRCA)
3. S R Pratibha, Anupaba Hongal “IOT Based Monitoring System in Smart Agriculture” 2017 International Conference on Recent Advances in Electronics and Communication Technology (ICRAECT)
4. <https://www.kaggle.com/datasets/atharvaingle/crop-recommendation-dataset>
5. Mohamed Rawidean, Mohd Kassim “IoT Applications in Smart Agriculture: Issues and Challenges” 2020 IEEE Conference on Open Systems (ICOS)
6. Iorliam, A., Bum, S., Aondoakaa, S. I., Iorliam I. B, &Shehu, Y. I. (2022).Machine Learning Techniques for the Classification of IoT-Enabled Smart IrrigationData for Agricultural Purposes.GU J Sci, Part A, 9(4), 378-391
7. HamzaBenyezz, MounirBouhedda, SamiaRebouh(2021). “Zoning irrigationsmart system based on fuzzy control technology and IoT for water and energy saving”. Journal of Cleaner Production Volume 302, 15 June 2021, 127001.
8. Reddy, BakkachennaRanadeep, et al. "Smart drip irrigation systemusingIOT."Proceedings of the International Conference on Innovative Computing&Communication (ICICC). 2021.
9. Ramya, S., A. M. Swetha, and M. Doraipandian. "IoT framework for smart irrigation using machine learning technique." Journal of Computer Science 16.3(2020): 355-363.
10. Sinwar et al. (2020) Sinwar D, Dhaka VS, Sharma MK, Rani G. Ai-based yieldprediction and smart irrigation. In: Pattnaik P, Kumar R, Pal S, editors. Internet of Things and Analytics for Agriculture, Volume 2—Studies in Big

Data. Vol. 67. Singapore: Springer; 2020. pp. 155–180.

11. Bhavsar, Drashti, et al. "A comprehensive and systematic study in smart dripandsprinkler irrigation systems." Smart Agricultural Technology 5 (2023): 100303.